

Robust Adaptive T -Count Lower Bounds for Boolean Oracles

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Abstract

Gosset, Kothari and Zhang (GKZ) showed that randomized Clifford+ T synthesis can sharply reduce the T -count of a Boolean oracle U_f , exponentially so for the multi-controlled Toffoli gate. For general f their lower bound for adaptive circuits, which may measure and act on the outcomes, holds only at zero error, and they left its extension to positive error as a challenge. We prove that for every non-affine f , an adaptive implementation within diamond error ε has worst-input expected T -count at least $(1 - 2\varepsilon/\delta_f)_+ \nu(U_f)$. Here ν is the stabilizer nullity and δ_f is the least distance from the affine functions of f and of its non-affine derivatives. For inner product on $2n$ bits this gives $\Theta(n)$ at every error up to $1/16$. For majority $\delta_f = \Theta(n^{-1/2})$, so the bound is empty at constant error. By counting the Pauli operators with large expectation in the Choi state, we show that majority nevertheless needs $\Theta(n)$ T gates at every error below an absolute constant. We also observe that GKZ's mixed-model lower bound applies to adaptive circuits whose feedforward consists of Pauli corrections. Finally, on the construction side, a sampled GKZ circuit that thresholds k parities can be run with $O(\log k)$ clean ancillae instead of $\Theta(k)$ at $O(k)$ T gates, and the multi-controlled Toffoli gate needs only one clean ancilla and $O(\min\{n, \log(2/\varepsilon)\})$ T gates.

Contents

1	Introduction	2
1.1	Our results	3
1.1.1	Implementations and costs	3
1.1.2	Lower bounds	3
1.1.3	Workspace for sampled circuits	4
1.2	Technical overview	5
1.3	Related work	7
1.4	Discussion and open problems	8
2	Preliminaries	9
2.1	Distances and synthesis models	9
2.2	Stabilizer and Choi states	11
3	Robust adaptive lower bounds	11
3.1	General oracles and the affine gap	11
3.1.1	Stabilizer nullity of Boolean oracles	11
3.1.2	The affine gap	12
3.1.3	Rigidity of branch nullity and an expected-cost lower bound	12
3.1.4	The exact bracket	13
3.2	Majority and symmetric Boolean oracles	15
3.2.1	All symmetric Boolean oracles	15
3.2.2	The majority gap and constant-error bounds	16

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4	Workspace for sampled circuits	17
4.1	Space-efficient evaluation of the GKZ top function	17
4.1.1	Fourier sampling produces a signed threshold	17
4.1.2	A blocked reversible tally	17
4.1.3	One clean ancilla with a read-only input	18
4.2	One-ancilla synthesis from affine-subspace tests	19
4.2.1	Randomized Toffoli from a linear-subspace test	19
4.2.2	Bounded spectral norm and a reusable clean ancilla	19
4.3	Memory lower bounds for threshold evaluators	20
4.3.1	Exact single-pass evaluation	20
4.3.2	Single-pass evaluation at bounded error	21
4.3.3	Transfer to sampled branches	21
4.3.4	Clean versus borrowed workspace	23
4.3.5	A pass–workspace hierarchy	24
A	Proofs of the lower bounds	26
A.1	Stabilizer nullity of Boolean oracles	26
A.2	The affine gap	26
A.3	Rigidity of branch nullity and an expected-cost lower bound	27
A.4	The exact bracket	28
A.5	All symmetric Boolean oracles	29
A.6	The majority gap and constant-error bounds	29
B	Binomial estimates for majority derivatives	31
C	Proofs of the workspace results	32
C.1	Fourier sampling produces a signed threshold	32
C.2	A blocked reversible tally	33
C.3	One clean ancilla with a read-only input	33
C.4	Randomized Toffoli from a linear-subspace test	35
C.5	Bounded spectral norm and a reusable clean ancilla	35
C.6	Exact single-pass evaluation	36
C.7	Single-pass evaluation at bounded error	36
C.8	Transfer to sampled branches	37
C.9	Clean versus borrowed workspace	37
C.10	A pass–workspace hierarchy	38

1 Introduction

A quantum algorithm evaluates a classical predicate $f : \mathbb{F}_2^n \rightarrow \{0, 1\}$ through the oracle $U_f|x\rangle|b\rangle = |x\rangle|b \oplus f(x)\rangle$, whether inside Grover search or a block encoding for the quantum singular value transform. In a surface-code architecture Clifford gates are comparatively cheap and every T gate consumes a distilled magic state [FG18, Lit19], so the number of T gates is the standard cost of such an oracle.

Gosset, Kothari and Zhang (GKZ) showed that randomization can lower this cost dramatically [GKZ25]. If each call to the oracle runs a Clifford+ T circuit sampled afresh, and only the averaged channel has to be close to U_f , then the n -qubit Toffoli gate needs $O(\log(1/\varepsilon))$ T gates at diamond error ε instead of $\Theta(n)$, and an arbitrary f needs $O(\|\widehat{f}\|_1^2 \log(1/\varepsilon))$, where $\|\widehat{f}\|_1$ is its Fourier spectral norm. GKZ also proved lower bounds. A random mixture of unitary circuits needs at least $\text{RPDT}_\varepsilon^{\text{na}}(f) - 1$ T gates, where $\text{RPDT}_\varepsilon^{\text{na}}$ is the randomized

non-adaptive parity decision tree complexity. Circuits that measure partway through and choose their later gates from the outcomes form a larger class, and for these GKZ prove a bound at $\varepsilon = 0$, and at positive error only for the Toffoli gate. A unitary channel is an extreme point of the set of channels, so an exact implementation must realize U_f on every branch, and randomization saves nothing until $\varepsilon > 0$. For general f they leave the extension of the adaptive bound to $\varepsilon > 0$ “as a challenge for future work”.

Our main result is such a bound for every Boolean oracle. For non-affine f we show

$$\mathcal{T}_\varepsilon^{\text{adaptive}}(U_f) \geq \left(1 - \frac{2\varepsilon}{\delta_f}\right)_+ \nu(U_f),$$

where $\mathcal{T}_\varepsilon^{\text{adaptive}}$ is the least worst-input expected T -count of an adaptive implementation within diamond error ε , $\nu(U_f)$ is the stabilizer nullity of the oracle, and δ_f , the affine gap, is the least distance from the affine functions among f and its non-affine derivatives (Definition 14). The nullity is $n + 1$ minus the dimension of the space of linear structures of f , so it is at most $n + 1$, and it is n or $n + 1$ for every symmetric non-affine f on $n \geq 3$ bits (Lemma 28). When the gap is bounded below by a constant, the bound is a constant fraction of $\nu(U_f)$ at constant error. This happens for every quadratic form, and inner product on $2n$ bits, whose oracle has nullity $2n + 1$, needs $\Theta(n)$ T gates at every $\varepsilon \leq 1/16$.

For majority the affine gap is $\Theta(n^{-1/2})$, because flipping one input changes the output only when the other bits are split evenly, so the bound above says nothing once $\varepsilon \gg n^{-1/2}$. Majority is the case we most wanted to settle, because it is the simplest symmetric function on which the gap bound fails and it is a threshold function, the same kind of function that every GKZ branch evaluates. Nothing suggests that majority is cheap, though, and we prove that it needs $\Theta(n)$ T gates at every error below an absolute constant (Theorem 2) by counting the Pauli operators with large expectation in the Choi state of U_{MAJ_n} . We also show that GKZ’s mixed-model bound applies without change to adaptive circuits whose feedforward consists of Pauli corrections (Proposition 3).

A second, smaller part of the paper concerns qubits. Written directly, a sampled GKZ circuit computes k parities of its input into k clean ancillae and thresholds them. We show that $O(\log k)$ clean ancillae suffice at the same asymptotic T -count, and that the Toffoli gate needs only one (Section 1.1.3).

1.1 Our results

1.1.1 Implementations and costs

A unitary implementation is a single circuit with clean ancillae that may be discarded at the end. A mixed implementation is a distribution over such circuits, and its cost is the largest T -count in the support. An adaptive implementation may measure and apply classical feedforward, and its cost is the worst-input expected number of T gates. A *branch* is what one run commits to: the sampled circuit in the mixed model, and the record of measurement outcomes and internal randomness in the adaptive one. The three costs satisfy

$$\mathcal{T}_\varepsilon^{\text{adaptive}}(U_f) \leq \mathcal{T}_\varepsilon^{\text{mixed}}(U_f) \leq \mathcal{T}_\varepsilon^{\text{unitary}}(U_f), \quad (1)$$

so a lower bound on $\mathcal{T}_\varepsilon^{\text{adaptive}}$ holds in all three models. Throughout, the error of an implementation $\mathcal{E}$ of a unitary U is the diamond distance $D_\diamond(\mathcal{E}, U) = \frac{1}{2} \|\mathcal{E} - \Phi_U\|_\diamond$, and Definitions 9 and 10 state the three costs formally.

1.1.2 Lower bounds

Theorem 1 (Adaptive lower bound from the affine gap (informal version of Theorem 17)). *For every non-affine f and every $\varepsilon \geq 0$,*

$$\mathcal{T}_\varepsilon^{\text{adaptive}}(U_f) \geq \left(1 - \frac{2\varepsilon}{\delta_f}\right)_+ \nu(U_f), \quad (x)_+ = \max\{x, 0\}.$$

In particular $\varepsilon \leq \delta_f/4$ forces $\mathcal{T}_\varepsilon^{\text{adaptive}}(U_f) \geq \frac{1}{2}\nu(U_f)$.

At $\varepsilon = 0$ this is the familiar nullity bound, which agrees with the parity decision tree bound of GKZ up to an additive one (Proposition 13). A branch that uses t T gates ends in a state of nullity at most t [BCHK20], and a branch whose nullity falls below $\nu(U_f)$ has fidelity at most $1 - \delta_f/2$ with the ideal Choi state, because each of its stabilizers either stabilizes that state or has expectation at most $1 - \delta_f$ on it. Error ε leaves such cheap branches a total probability of at most $2\varepsilon/\delta_f$, and working with stabilizer projectors rather than trace distance keeps the loss linear in ε/δ_f .

For AND_m the gap is 2^{-m} , and the theorem recovers GKZ's bound $\Omega(\min\{n, \log(1/\varepsilon)\})$ for the Toffoli gate (Example 18). For a quadratic f whose alternating form has rank $2h$, the gap is $\frac{1}{2}(1 - 2^{-h}) \geq \frac{1}{4}$ and the nullity is $2h + 1$, so $\mathcal{T}_\varepsilon^{\text{adaptive}}(U_f) \geq h$ for every $\varepsilon \leq 1/16$ (Corollary 23). Inner product on $2n$ bits has $h = n$. In the mixed model an $\Omega(n)$ bound for it already follows from GKZ and the communication complexity of inner product (Remark 24).

Theorem 2 (Majority at constant error (informal version of Theorem 33)). *There is an absolute constant $\varepsilon_* > 0$ such that*

$$\mathcal{T}_\varepsilon^{\text{adaptive}}(U_{\text{MAJ}_n}) = \Theta(n)$$

for every odd $n \geq 5$ and every $0 \leq \varepsilon \leq \varepsilon_$. The same holds in the mixed and unitary models, by (1) and exact synthesis.*

Majority is far from affine, and its gap is small only because a single-coordinate derivative of MAJ_n is within $O(n^{-1/2})$ of the zero function (Proposition 32). The proof bounds the fidelity of a low-nullity state with the Choi state by the number K_τ of Pauli operators whose expectation there exceeds $1 - \tau$ (Lemma 19). For majority these come from derivatives $\partial_S \text{MAJ}_n$ that are nearly constant, which happens only for shifts S of weight below $n/4$ or above $3n/4$, and there are too few such shifts for a state of nullity below αn , for a fixed $\alpha > 0$, to approximate the Choi state. The constant ε_* is not explicit (it comes from an anticoncentration bound), and in the mixed model an $\Omega(n)$ bound for majority already appears in Table 1 of GKZ [GKZ25].

Proposition 3 (Pauli-frame circuits (informal version of Proposition 26)). *Suppose an adaptive Clifford+ T circuit measures single qubits in the computational basis, places its t T gates independently of the outcomes, and its only classically controlled gates are Paulis applied according to affine functions of the outcomes. If it implements U_f within diamond error ε , then $t \geq \text{RPDT}_\varepsilon^{\text{na}}(f) - 1$.*

The proof defers the measurements. A Pauli controlled by an affine function of recorded outcomes is a product of controlled Paulis, hence Clifford, so the deferred circuit is a mixed implementation with the same T gates and GKZ's mixed-model bound applies. Clifford or T corrections, and corrections controlled by nonlinear functions of the outcomes, add a term proportional to their number and to the multiplicative complexity of their conditions (Proposition 26). (Measurement-based uncomputation of an AND gate applies a CZ correction, so it pays this term once per gadget.)

1.1.3 Workspace for sampled circuits

GKZ's general construction, which we call the *Fourier sampler*, draws k Fourier characters of f from the distribution $|\widehat{f}(S)|/L$, where $L = \|\widehat{f}\|_1$, and outputs a circuit that computes the corresponding k parities of the input and applies a fixed threshold to them, the branch's *top function*. For nonzero f at error ε ,

$$k = \Theta\left(L^2 \log(8/\varepsilon)\right) \quad (2)$$

(Section 4.1.1). Parities need only CNOT gates, so the top function accounts for all of a branch's T gates and all of its workspace. We count λ , the clean ancillae a circuit allocates beyond its input and target, and let a circuit borrow a data qubit for free if it restores it exactly. Like GKZ, we assume the distribution $|\widehat{f}(S)|/L$ can be sampled, which for a general f given as a circuit can take time exponential in n .

Proposition 4 (Workspace-efficient tally (informal version of Proposition 35)). *Every nonzero Boolean oracle U_f has a mixed implementation at diamond error ε with*

$$T = O(k) \quad \text{and} \quad \lambda = O(\log k),$$

for every $0 < \varepsilon < 1$.

The circuit counts the parities a block of $\Theta(\log k)$ at a time, adds each block's count into an accumulator, and recomputes the block's parities to clear it before starting the next, in the manner of the logarithmic-workspace Hamming-weight circuits of Bravyi, Yoder and Maslov and of Zi et al. [BYM22, ZNS25].

Proposition 5 (Toffoli with one clean ancilla (informal version of Proposition 41)). *For every $n \geq 2$ and $0 < \varepsilon < 1$, Toff_n has a mixed implementation with at most one clean ancilla and $O(\min\{n, \log(2/\varepsilon)\})$ T gates per branch, and every work qubit is restored exactly. When $\lceil \log(4/\varepsilon) \rceil \leq n - 2$ no clean ancilla is needed.*

GKZ's Toffoli sampler tests whether $\ell = \lceil \log(4/\varepsilon) \rceil$ random parities of the controls all vanish. After a CNOT change of coordinates this test is a single ℓ -controlled gate, and Barenco et al. implement that gate with one borrowed work qubit [BBC⁺95].

Section 4 also has a coherent version of Barrington's theorem that evaluates a branch with the target and one further qubit at polynomial T -count without writing to the input (Proposition 36), one qubit fewer than Ben-Or and Cleve's exact three-register programs [BOC92] at the cost of exactness. Every f of spectral norm at most M has a mixed implementation with one clean ancilla and $O_M(\log(2/\varepsilon))$ T gates, uniformly in n , through the Green–Sanders decomposition into coset indicators (Proposition 43). Its dependence on M is exponential, and the proof does not construct the decomposition. The same section proves memory lower bounds for evaluators that read the k threshold bits in a fixed order (Table 1), which bound an evaluator's total memory w and say nothing directly about λ .

Task or model	Bound	Scope
Non-affine f	$\mathcal{T}_\varepsilon^{\text{adaptive}} \geq (1 - 2\varepsilon/\delta_f)_{+} \nu(U_f)$	Every $\varepsilon \geq 0$ (Theorem 17)
Quadratic f , rank $2h$	$\mathcal{T}_\varepsilon^{\text{adaptive}} \geq h$	$\varepsilon \leq 1/16$ (Corollary 23)
Odd majority	$\mathcal{T}_\varepsilon^{\text{adaptive}} = \Theta(n)$	$0 \leq \varepsilon \leq \varepsilon_*$ (Theorem 33)
Pauli-frame circuits	$t \geq \text{RPDT}_\varepsilon^{\text{na}}(f) - 1$	Every $\varepsilon \geq 0$ (Proposition 26)
Fourier branch	$\lambda = O(\log k), T = O(k)$	Mixed oracle (Proposition 35)
Toff_n	$\lambda \leq 1, T = O(\min\{n, \log(2/\varepsilon)\})$	$\lambda = 0$ when $\ell \leq n - 2$ (Proposition 41)
$\ \widehat{f}\ _1 \leq M$	$\lambda \leq 1, T = O_M(\log(2/\varepsilon))$	Existential (Proposition 43)
Threshold evaluators	$w = \Omega(\log k), wp = \Omega(\log k),$ $D \geq k + 1$	One pass; p passes; commuting unitaries (Section 4.3)

Table 1: Summary of results. Lower bounds on T -count are above the rule and workspace results below it. Here $\ell = \lceil \log(4/\varepsilon) \rceil$, w is an evaluator's memory in qubits and D its dimension, and O_M hides a constant depending on M alone.

1.2 Technical overview

The affine gap (Theorem 17). Run an adaptive circuit on half of a maximally entangled pair, a stabilizer state, so that the ideal output is the Choi state of U_f . A branch using t_b T gates ends in a pure state $|\Psi_b\rangle$ on the Choi registers and the retained ancillae, of nullity at most t_b . If F_b is the fidelity of its reduced state with the ideal Choi state $|J_{U_f}\rangle$, the error bound gives $\sum_b p_b(1 - F_b) \leq \varepsilon$. For each stabilizer $P = \pm Q \otimes R$ of $|\Psi_b\rangle$, split across the Choi registers and the ancillae, Uhlmann's theorem and the projector $(I + P)/2$ give

$$F_b \leq \frac{1 + |\langle J_{U_f} | Q | J_{U_f} \rangle|}{2}.$$

Every Pauli expectation of $|J_{U_f}\rangle$ is an average of two correlations of affine characters with f , a shift of f , or a derivative of f . Each Pauli therefore either stabilizes $|J_{U_f}\rangle$ or has expectation at most $1 - \delta_f$ there (Lemma 15). Fidelity above $1 - \delta_f/2$ forces every projected stabilizer of the branch to stabilize the ideal Choi state. Such a branch has nullity at least $\nu(U_f)$. The ancillae cannot prevent this, because r retained ancillae absorb at most 2^r stabilizers per Choi Pauli. Branches with less nullity have infidelity at least $\delta_f/2$ and total probability at most $2\varepsilon/\delta_f$.

Majority (Theorem 33). The gap argument used only the largest nontrivial Pauli expectation. Using the whole stabilizer group of a branch gives more. If $K_\tau(J)$ counts the N -qubit Pauli expectations of $|J\rangle$ with magnitude above $1 - \tau$, then any purification of nullity ν satisfies

$$F \leq 1 - \tau + \tau \min\{1, K_\tau(J)2^{\nu-N}\},$$

since at most $2^r K_\tau$ of the $2^{N+r-\nu}$ stabilizers can have large expectation. For majority the only potentially large coefficients are p_S and $1 - p_S$, where $p_S = \Pr[\partial_S \text{MAJ}_n = 1]$. Both are bounded away from zero for shifts of weight between $n/4$ and $3n/4$ (Lemma 31). The remaining shifts lie in binomial tails, which gives $K_\tau(J) \leq 2^{N-\alpha n}$ for $\alpha = 1 - h_2(1/4) > 0$ and an absolute $\tau > 0$. Branches of nullity below $\alpha n - 1$ have constant infidelity, and their total probability is $O(\varepsilon)$. This proves the bound for $n \geq 11$. For $n = 5, 7, 9$ the gap bound applies directly, and ε_* is the smaller of the two thresholds.

Pauli-frame circuits (Proposition 26). Replace each measurement by a CNOT onto a fresh record qubit. A Pauli applied when an affine function of the record equals one becomes a product of Paulis controlled by record qubits, which is Clifford. Dephasing the record qubits before they are discarded leaves the reduced channel unchanged. The result is a mixed implementation with the same t T gates, to which GKZ's bound $\mathcal{T}_\varepsilon^{\text{mixed}}(U_f) \geq \text{RPDT}_\varepsilon^{\text{na}}(f) - 1$ applies. Nonlinear conditions and non-Pauli corrections are computed reversibly from the record, and their T -count is the additive term.

Workspace constructions (Propositions 35 and 41). A sampled signed character $\sigma \chi_S(x)$ has expectation $f(x)/L$. Writing its sign as a complement of the parity bit turns the rounding rule of the Fourier sampler into $P \leq \theta$, where P counts the parities that are set and $k/4 - 1 \leq \theta < k/2$. Incrementing an accumulator once per parity would cost $O(k \log k)$ Toffoli gates. Instead, the tally computes a block of $\beta = \Theta(\log k)$ parities into clean scratch, counts them with a tree of small adders, adds the count into an $O(\log k)$ -qubit accumulator, and uncomputes the block (Figure 1). Each of the $O(k/\log k)$ blocks costs $O(\log k)$ Toffolis. The accumulation therefore costs $O(k)$, and the comparison and final uncomputation cost no more.

For Toff $_n$ the GKZ sampler computes an OR of ℓ random parities, which misses a nonzero input with probability at most $2^{-\ell}$. Choosing linearly independent parity rows lets a CNOT basis change turn the branch into one ℓ -controlled gate (Figure 2). Barenco's decomposition of that gate uses one additional work qubit, which can be an unused data coordinate if one is available [BBC⁺95].

For bounded spectral norm, the Green–Sanders theorem writes f as a signed sum of at most $q(M)$ coset indicators [GS08, San19]. Modulo two the signs disappear, each coset indicator is a Clifford-conjugated Toffoli, and applying Proposition 41 to each at error $\varepsilon/q(M)$ gives Proposition 43. The Barrington construction of Proposition 36 runs a width-five permutation branching program through a four-dimensional orthogonal representation of A_5 on the target and one ancilla.

Memory bounds (Section 4.3). In one pass the workspace is everything the suffix learns about the prefix. It must therefore separate prefixes of different weights, and Nayak's bound for serial encodings gives $\Omega(\log k)$ qubits at every fixed error below $1/2$ [Nay99]. With several passes, Alice and Bob simulate the reads on either side of a balanced cut and send the workspace at each of the c crossings, and the rank bound for quantum communication [Kre95, BdW01] against a threshold staircase of rank $\Omega(k)$ gives $wc = \Omega(\log k)$. With commuting

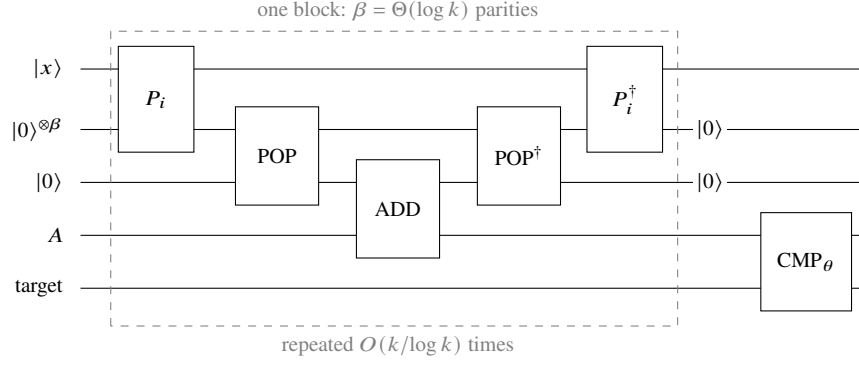


Figure 1: A block of the reversible tally. Parities are computed into clean scratch (P_i) and counted into a block-sum register (POP), the count is added to the accumulator A (ADD), and the block is uncomputed. The comparison CMP_θ runs after all blocks, and reversing the accumulation clears the remaining workspace.

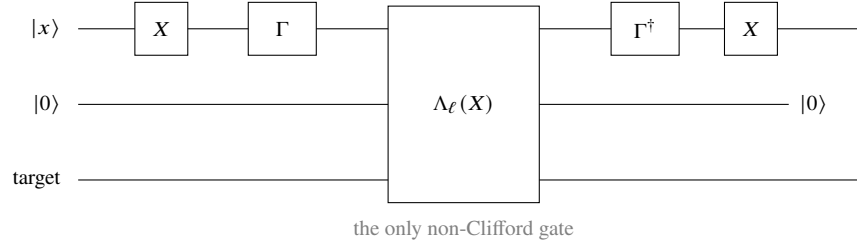


Figure 2: A branch of the Toff_n implementation of Proposition 41. Γ is the CNOT basis change and $\Lambda_\ell(X)$ the ℓ -fold controlled NOT with negative controls. The lone ancilla is its work qubit, returned to $|0\rangle$.

unitary updates, exactness forces the $k + 1$ states obtained by applying the updates along a fixed ordering to be mutually orthogonal.

1.3 Related work

Stabilizer nullity is the standard tool for T -count lower bounds. Beverland et al. showed that it cannot increase under postselected stabilizer operations and grows by at most one per T gate [BCHK20], so it survives measurement and feedforward, and it has since been applied to unitaries and to state preparation [JW21, GKW24]. For Boolean oracles, Gosset, Kothari and Zhang prove $\mathcal{T}_\epsilon^{\text{mixed}}(U_f) \geq \text{RPDT}_\epsilon^{\text{na}}(f) - 1$ in the mixed model at every error, and an adaptive bound by parity decision trees at zero error [GKZ25]. Our nullity bound agrees with the latter up to an additive constant (Proposition 13). Other monotones, among them stabilizer extent, robustness of magic and the channel monotones of Wang, Wilde and Su, bound T -counts through their growth per T gate [BBC⁺19, HC17, WWS19], but like nullity they measure a single output state, whereas our bounds use how the error is spread over the branches. A comparison of stabilizer fidelities cannot give a growing bound for our oracles, since every Boolean-oracle Choi state has fidelity $1/2$ with a stabilizer state, obtained by putting both target qubits in $|+\rangle$.

The closest technique we know of is due to Luo and Li, who bound the rank of stabilizer subgroups whose projector retains weight on states near W states and obtain worst-case T -count bounds for approximate sparse-state preparation [LL26]. Lemma 16 applies a similar projection to Choi states of oracles and averages it over the branches of an adaptive implementation. Counting Pauli operators with large expectation is familiar from stabilizer testing and from pseudorandomness bounds for states prepared with few T gates [GNW21, GIKL23]. Girish et al. and Parham bound magic through communication complexity [GMPY25, Par25], and Remark 24

compares that route with ours on inner product and majority.

On the construction side, Low, Kliuchnikov and Schaeffer prepare an arbitrary N -amplitude state from $O(\log(N/\varepsilon))$ clean qubits and roughly $\mu \log((\log N)/\varepsilon)$ dirty qubits, for a tunable parameter μ , at T -count $O(N/\mu + \mu \log(N/\varepsilon) \log((\log N)/\varepsilon))$, and show by counting that this is optimal up to logarithmic factors [LKS24]. Theirs is the closest tradeoff between clean workspace and T -count that we know of, though for state preparation rather than oracles. Selinger, and Amy and Mosca, minimize the T -depth and T -count of exact circuits [Sel13, AM19], and more recent work covers Boolean oracles and near-Clifford unitaries [NZ26, DBCM25, FHW26], all in the exact setting where randomization saves nothing. Dirty and conditionally clean ancillae [BBC⁺95, HRS16, Gid17, TT16, KG24], like catalytic computation [BCK⁺14, BFM⁺25], replace clean workspace by borrowed qubits that are returned intact, and our constructions borrow data qubits in this sense. For the Toffoli gate, Koe et al. reach one clean ancilla with mid-circuit measurement [KASD26, KSD26]. Their decompositions are exact, so GKZ’s zero-error adaptive bound makes their T -count at least linear in n , whereas Proposition 41 uses $O(\min\{n, \log(2/\varepsilon)\})$ T gates without measurement. Yamazaki and Akibue show that $3 \log_2(1/\varepsilon) + o(\log(1/\varepsilon))$ T gates suffice for most multi-qubit controlled $SU(2)$ gates [YA26], a count that, like the one in Proposition 41, does not grow with the number of controls. Zhang et al. show that adaptivity removes the logarithmic overhead of compiling arbitrary one- and two-qubit gates to Clifford+ T [?], whereas our lower bounds limit what adaptivity can save for Boolean oracles.

The tally of Proposition 35 is ordinary reversible arithmetic [Gid18, Dra00]. Bravyi, Yoder and Maslov compute a Hamming weight with $O(\log n)$ ancillae and $O(n \log n)$ gates, and Zi and coauthors study the tradeoff between depth and ancillae for Hamming weight and symmetric functions [BYM22, ZNS25, ZYNZ26]. The linear gate count implicit in GKZ comes from the linear multiplicative complexity of threshold functions [BPP00], which also controls exact oracle synthesis [MSC⁺19]. Green and Sanders give the coset decomposition behind Proposition 43 [GS08], and Sanders shortens it to $\exp(M^{4+o(1)})$ terms in every finite abelian group [San20] and to $\exp(M^{3+o(1)})$ on $\mathbb{F}_2^n$ [San19].

The evaluator bounds of Section 4.3 use Nayak’s serial encoding and the rank method for quantum communication [Nay99, ANTSV02, Kre95, BdW01], and the evaluators themselves are close relatives of quantum branching programs and finite automata [SS05, BP99]. Length lower bounds for bounded-width branching programs [AM86, BPRS90, BST90] are discussed in Section 1.4. At constant width the reference point is Barrington’s theorem [Bar89], with algebraic and quantum versions due to Ben-Or and Cleve and to Ablayev et al. [BOC92, AGK⁺05]. Kornerup, Sadun and Soloveichik trade workspace against time in a pebbling model [KSS25], where we trade it against T -count.

1.4 Discussion and open problems

Theorem 33 settles the adaptive T -count of majority at sufficiently small constant error, but ε_* is not explicit and the range between ε_* and $1/2$ is open. The counting argument is not specific to majority and may give expected-cost lower bounds for other functions whose gap is small only because a few derivatives are nearly constant. In the mixed model GKZ’s bound $\text{RPDT}_\varepsilon^{\text{na}}(f) - 1$ holds at every error, and Proposition 26 carries it to Pauli-frame circuits. Whether it holds for all adaptive circuits is open.

Open Problem 6 (Adaptive circuits and parity decision trees). Is there, for every fixed $\varepsilon < 1/2$, a constant $c_\varepsilon > 0$ such that $\mathcal{T}_\varepsilon^{\text{adaptive}}(U_f) \geq c_\varepsilon \text{RPDT}_\varepsilon^{\text{na}}(f) - 1$ for every f ?

The additive term cannot be dropped, since a nonconstant affine f has $\text{RPDT}_\varepsilon^{\text{na}}(f) = 1$ and $\mathcal{T}_\varepsilon^{\text{adaptive}}(U_f) = 0$. At $\varepsilon = 0$ the inequality holds with $c_0 = 1$ by Proposition 13, and Theorems 17 and 33 give it for quadratic forms and majority at small constant error.

On the workspace side, Proposition 43 gives one clean ancilla for every oracle of bounded spectral norm, at a T -count independent of n but exponential in M .

Open Problem 7 (Polynomial dependence on the spectral norm). Is there a single polynomial P such that every f with $\|\hat{f}\|_1 \leq M$ has a mixed implementation of U_f at diamond error ε with at most $P(M) \log(8/\varepsilon)$ T gates

and constant clean workspace, with data borrowing allowed?

Sanders' decomposition has $\exp(M^{3+o(1)})$ terms [San19], so a polynomial bound would need either a much shorter decomposition into coset indicators or a synthesis route that does not pass through one. At the branch level the question is sharper.

Open Problem 8 (Linear T -count for a read-only threshold branch). Is there an absolute constant C such that, for every $k \geq 16$ and every integer $k/4 - 1 \leq \theta \leq 3k/4$, the coherent oracle U_{THR_θ} has an exact Clifford+ T circuit with at most C clean ancillae and at most CkT gates, using each input qubit only as a control and returning all ancillae to $|0\rangle$? The input may be reread arbitrarily often.

Each budget is attainable separately, since the tally of Proposition 35 meets the T -count and the three-register programs of Ben-Or and Cleve [BOC92] meet the workspace, with one clean and one borrowed qubit or with two borrowed qubits (Section 4.1.3). Proposition 39 rules out exact synthesis with no ancilla at all in the most restricted read-only gate model, which is why the question allows a constant. Classical length lower bounds for bounded-width branching programs [AM86, BPRS90] bound the length of a program of fixed width and count every step, whereas a T -count counts only the non-Clifford ones, so they do not settle it.

2 Preliminaries

All logarithms are base 2 unless written as $\ln$, and $[n] = \{1, \dots, n\}$. We identify a subset $S \subseteq [n]$ with its characteristic vector in $\mathbb{F}_2^n$, so that S is at once a set and a vector, and $|S|$ is the Hamming weight of that vector. $\mathbf{1}$ is the all-ones vector, the characteristic vector of $[n]$ itself; $\mathbf{1}_A$ is the indicator function of a set A ; and $\mathbf{1}[P]$ is 1 when the predicate P holds and 0 otherwise.

For $S, x \in \mathbb{F}_2^n$, write $\langle S, x \rangle$ for their mod-2 inner product, the *parity* of the bits of x that S selects, and $\chi_S(x) = (-1)^{\langle S, x \rangle}$. A Boolean function is regarded as $\{0, 1\}$ -valued in $\mathbb{R}$ whenever its Fourier coefficients are taken, and the normalized expansion of $f : \mathbb{F}_2^n \rightarrow \mathbb{R}$ is

$$f(x) = \sum_{S \in \mathbb{F}_2^n} \widehat{f}(S) \chi_S(x), \quad \widehat{f}(S) = 2^{-n} \sum_x f(x) \chi_S(x),$$

with $\|\widehat{f}\|_1 = \sum_S |\widehat{f}(S)|$. We write $\text{AND}_m(x_1, \dots, x_m) = x_1 \wedge \dots \wedge x_m$, and Aff for the affine functions $\mathbb{F}_2^n \rightarrow \mathbb{F}_2$, that is the parities $x \mapsto \langle a, x \rangle \oplus c$.

For a Boolean function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, its oracle unitary acts on n input qubits and one target qubit as $U_f|x\rangle|b\rangle = |x\rangle|b \oplus f(x)\rangle$. The n -qubit Toffoli gate is $\text{Toff}_n = U_{\text{AND}_{n-1}}$, with $n-1$ controls.

Fix vectors $S_1, \dots, S_q$ in advance. A *non-adaptive parity decision tree* asks for the q parities $\langle S_1, x \rangle, \dots, \langle S_q, x \rangle$ and returns an output bit determined by those q answers alone; the number of questions q is its *depth*. Because the S_i are chosen before any answer comes back, no question depends on an earlier one, so the tree amounts to a fixed list of q questions together with a rule sending the q answer bits to one output bit. Section 3.2 also calls such a tree a sketch.

Write $\text{PDT}^{\text{na}}(f)$ for the least depth of such a tree that returns $f(x)$ on every input x . For the randomized version, call a probability distribution over these trees ε -correct if on every input it returns the right value with probability at least $1 - \varepsilon$, and write $\text{RPDT}_\varepsilon^{\text{na}}(f)$ for the least q for which some ε -correct distribution uses only trees of depth at most q . The cost is the deepest tree the distribution can produce, not the average depth. Gosset, Kothari and Zhang bound the T -count of an arbitrary Boolean oracle in these terms, as $\mathcal{T}_\varepsilon^{\text{mixed}}(U_f) \geq \text{RPDT}_\varepsilon^{\text{na}}(f) - 1$ (Definition 9 fixes $\mathcal{T}_\varepsilon^{\text{mixed}}$) [GKZ25].

2.1 Distances and synthesis models

For states ρ, σ , let $D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1$, where $\|A\|_1 = \text{Tr} \sqrt{A^\dagger A}$ is the trace norm. A single two-outcome measurement distinguishes ρ from σ with success probability at most $\frac{1}{2}(1 + D(\rho, \sigma))$ when each is given with

probability $1/2$, and some measurement attains it. A subscript 1 means the trace norm on an operator and the Fourier ℓ^1 norm on a function, as in $\|\widehat{f}\|_1$ above. We also write $\|A\|_\infty$, or $\|A\|_{\text{op}}$ where singular values are in play, for the operator norm, the largest singular value of A ; $\sigma_{\min}(A)$ is the smallest, and $\|A\|_F$ is the Frobenius norm, the square root of the sum of the squared entries. For channels $\mathcal{E}, \mathcal{F}$, let

$$D_\diamond(\mathcal{E}, \mathcal{F}) = \frac{1}{2} \|\mathcal{E} - \mathcal{F}\|_\diamond = \max_\rho D((\mathcal{E} \otimes I)(\rho), (\mathcal{F} \otimes I)(\rho)),$$

the maximum running over inputs ρ on the channel's own system together with a reference system that neither channel touches. We abbreviate $D_\diamond(\mathcal{E}, U) = D_\diamond(\mathcal{E}, \Phi_U)$, where $\Phi_U(\rho) = U\rho U^\dagger$.

Definition 9 (Unitary and mixed T -count). A unitary Clifford+ T implementation of U is a single Clifford+ T circuit acting on the input and ancillae initialized to $|0\rangle$, after which the ancillae are discarded. $\mathcal{T}_\varepsilon^{\text{unitary}}(U)$ is the minimum number of T gates in such a circuit whose channel is within diamond distance ε of U .

A mixed implementation is a probability distribution over such circuits. One circuit drawn from it is a branch, and the cost is the largest number of T gates on any branch. The minimum cost at diamond error ε is $\mathcal{T}_\varepsilon^{\text{mixed}}(U)$.

Definition 10 (Adaptive expected T -count). An adaptive Clifford+ T circuit is as in Definition 9, except that it may also use classical randomness, measure qubits in the computational basis, and choose its later gates from the outcomes. Here a branch b is the record of those outcomes together with the internal randomness, and t_b is the number of T gates it uses. For a pure input $|\psi\rangle$ on the input and target registers, let $t(\psi)$ be the expectation of t_b over the branch distribution that $|\psi\rangle$ induces, and let the cost of the circuit be $\sup_\psi t(\psi)$. Since t is affine in the input state, that supremum also dominates the cost on any mixed input, and on any input entangled with a reference system the circuit does not touch. Then $\mathcal{T}_\varepsilon^{\text{adaptive}}(U)$ is the least cost among such circuits whose induced channel is within diamond distance ε of U .

These give the inclusions (1). A unitary implementation is a mixed one supported on a single circuit, so $\mathcal{T}_\varepsilon^{\text{mixed}} \leq \mathcal{T}_\varepsilon^{\text{unitary}}$. A mixed implementation is an adaptive circuit that uses only its classical randomness, and its expected T -count is at most the largest count in the support, so $\mathcal{T}_\varepsilon^{\text{adaptive}} \leq \mathcal{T}_\varepsilon^{\text{mixed}}$. In an exact adaptive implementation, every nonzero-probability branch produces the ideal pure output, so a branch lower bound automatically controls the expectation. At positive error a lower bound must also show that expensive branches carry substantial probability, and the averaging argument in Theorem 17 supplies this.

For mixed implementations, λ denotes the largest number of clean ancilla qubits any branch holds simultaneously at any one time, excluding the input qubits and target. A branch may touch far more clean qubits in total than λ , provided it clears and reuses them. Data qubits may be modified temporarily only if they are restored exactly.

Lemma 11 (Error from pointwise miss probabilities). *Let U be a unitary and let a random unitary branch be $U(I - 2\Pi_a)$, where Π_a is a projector. If*

$$\|\mathbb{E}_a \Pi_a\|_\infty \leq p,$$

then the averaged channel is within diamond distance $4p$ of U . In particular, for any Boolean f and any distribution of Boolean functions g_a , if $q = \max_x \Pr[g_a(x) \neq f(x)]$, then

$$D_\diamond(\mathbb{E}_a \Phi_{U_{g_a}}, \Phi_{U_f}) \leq 4q.$$

Conditioning the branch distribution on an event E of positive probability gives the bound $4q/\Pr[E]$.

Proof. Conjugating by U does not affect the distance. For a pure input state ψ on the system and an arbitrary reference, put $\bar{\Pi} = \mathbb{E}_a \Pi_a$. Expanding gives

$$\mathbb{E}_a [(I - 2\Pi_a)\psi(I - 2\Pi_a)] - \psi = -2\bar{\Pi}\psi - 2\psi\bar{\Pi} + 4\mathbb{E}_a [\Pi_a\psi\Pi_a].$$

The first two terms have total trace norm at most $4 \left\| \overline{\Pi} \right\|_\infty$. The last is positive and has trace norm

$$4\mathbb{E}_a \text{Tr}(\Pi_a \psi) = 4 \text{Tr}(\overline{\Pi} \psi) \leq 4 \left\| \overline{\Pi} \right\|_\infty.$$

After the factor $1/2$ in trace distance, the result is at most $4p$. Taking the supremum over the reference and pure inputs proves the diamond bound. For Boolean oracles, use

$$\Pi_a = \sum_{x: g_a(x) \neq f(x)} |x\rangle\langle x| \otimes |-\rangle\langle -|, \quad U_{g_a} = U_f(I - 2\Pi_a).$$

Then $\|\mathbb{E}_a \Pi_a\|_\infty = q$. After conditioning, each input's miss probability is at most $q/\Pr[E]$, which proves the last assertion. $\square$

2.2 Stabilizer and Choi states

A *stabilizer state* is one obtained from $|0\rangle^{\otimes q}$ by a Clifford circuit. Equivalently, it is the unique joint $+1$ eigenstate of a maximal abelian subgroup (*stabilizer*) of the Pauli group that does not contain $-I$.

Feed a q -qubit channel $\mathcal{E}$ half of the maximally entangled state

$$|\Omega_q\rangle = 2^{-q/2} \sum_{x \in \mathbb{F}_2^q} |x\rangle |x\rangle.$$

What comes out is the *Choi state*

$$J(\mathcal{E}) = (\mathcal{E} \otimes I)(|\Omega_q\rangle\langle\Omega_q|),$$

a single state on $2q$ qubits that determines $\mathcal{E}$ completely. For a unitary the Choi state is pure, $|J_U\rangle = (U \otimes I)|\Omega_q\rangle$, and for an oracle U_f we have $q = n + 1$, so $|J_{U_f}\rangle$ is a state on $N = 2(n + 1)$ qubits. Since $|\Omega_q\rangle$ is prepared from $|0\rangle^{\otimes 2q}$ by Hadamards and CNOTs, it is itself a stabilizer state, so the adaptive bounds of [Section 3.1](#) can start from it.

If $D_\diamond(\mathcal{E}, U) \leq \varepsilon$, then

$$D(J(\mathcal{E}), |J_U\rangle\langle J_U|) \leq \varepsilon \quad \text{and hence} \quad \langle J_U | J(\mathcal{E}) | J_U \rangle \geq 1 - \varepsilon. \quad (3)$$

The first inequality is the case $\rho = |\Omega_q\rangle\langle\Omega_q|$ of the maximum defining $D_\diamond$, and the second follows by measuring the two-outcome POVM $\{|J_U\rangle\langle J_U|, I - |J_U\rangle\langle J_U|\}$.

3 Robust adaptive lower bounds

Proofs of the longer results are in [Appendix A](#).

3.1 General oracles and the affine gap

3.1.1 Stabilizer nullity of Boolean oracles

Let $\mathcal{P}_N$ denote the N -qubit Pauli group modulo global phase. Each element has two Hermitian representatives, P and $-P$, and at most one of them can fix a given state. For a pure N -qubit state $|\psi\rangle$ we therefore count an element as stabilizing when either representative fixes $|\psi\rangle$:

$$\text{Stab}(\psi) = \{P \in \mathcal{P}_N : P|\psi\rangle = |\psi\rangle \text{ or } -P|\psi\rangle = |\psi\rangle\}, \quad \nu(\psi) = N - \log |\text{Stab}(\psi)|.$$

$\text{Stab}(\psi)$ is a subgroup, so its order is a power of two and the *stabilizer nullity* $\nu(\psi)$ is a non-negative integer. For a q -qubit unitary U , define

$$C(U) = \{P \in \mathcal{P}_q : UPU^\dagger \in \mathcal{P}_q\}, \quad \nu(U) = 2q - \log |C(U)|.$$

The Choi state satisfies $\nu(J_U) = \nu(U)$. Stabilizer nullity is additive under tensor products, does not increase under Clifford operations or postselected Pauli measurements, and increases by at most one per T gate [BCHK20].

For $S \in \mathbb{F}_2^n$, define the discrete derivative $\partial_S f(x) = f(x) \oplus f(x \oplus S)$ and the space of linear structures

$$L(f) = \{S \in \mathbb{F}_2^n : \partial_S f \text{ is constant}\}.$$

Lemma 12 (Oracle nullity). *If f is affine, then $\nu(U_f) = 0$. If f is non-affine, then*

$$\nu(U_f) = n + 1 - \dim L(f).$$

For non-affine functions the exact nullity of Lemma 12 agrees, up to one, with GKZ's non-adaptive parity-decision-tree bound.

Proposition 13 (Nullity versus parity decision trees). *Let f be non-affine and let*

$$I(f) = \{S \in \mathbb{F}_2^n : \partial_S f \equiv 0\}.$$

Then

$$\text{PDT}^{\text{na}}(f) = n - \dim I(f), \quad \nu(U_f) \in \{\text{PDT}^{\text{na}}(f), \text{PDT}^{\text{na}}(f) + 1\}.$$

Proof. A collection of parity queries spanning a subspace V computes f exactly iff f is constant on cosets of $V^\perp$, equivalently $V^\perp \subseteq I(f)$. The minimum possible $\dim V$ is therefore $n - \dim I(f)$.

The map $L(f) \rightarrow \mathbb{F}_2$ sending S to the constant value of $\partial_S f$ is a homomorphism with kernel $I(f)$. Hence $I(f)$ has index one or two in $L(f)$. Substitute this into Lemma 12. $\square$

3.1.2 The affine gap

For a Boolean function g , let

$$\text{dist}(g, \text{Aff}) = \min_{\ell \in \text{Aff}} 2^{-n} |\{x : g(x) \neq \ell(x)\}|.$$

Definition 14 (Affine gap). For non-affine f , define

$$\delta_f = \min \left\{ \text{dist}(f, \text{Aff}), \min_{S: \partial_S f \notin \text{Aff}} \text{dist}(\partial_S f, \text{Aff}) \right\},$$

where the inner minimum is $+\infty$ if every derivative is affine. Then $0 < \delta_f \leq 1/2$.

In the language of cryptographic Boolean functions, $\text{dist}(g, \text{Aff})$ is the nonlinearity of g divided by 2^n , and the space $L(f)$ of Lemma 12 is the space of linear structures of f .

Lemma 15 (Pauli gap). *For every Pauli Q on the $2(n+1)$ qubits of $|J_{U_f}\rangle$, either a Hermitian representative of Q stabilizes $|J_{U_f}\rangle$, or*

$$|\langle J_{U_f} | Q | J_{U_f} \rangle| \leq 1 - \delta_f.$$

3.1.3 Rigidity of branch nullity and an expected-cost lower bound

Lemma 16 (Nullity rigidity near the Choi state). *Let $|\Psi\rangle$ be a pure state on the Choi registers and r additional qubits, and write Tr_{anc} for the partial trace over those r qubits, so that $\rho = \text{Tr}_{\text{anc}} |\Psi\rangle\langle\Psi|$ is its reduced state on the Choi registers. If*

$$F = \langle J_{U_f} | \rho | J_{U_f} \rangle > 1 - \frac{\delta_f}{2},$$

then

$$\nu(\Psi) \geq \nu(U_f).$$

Theorem 17 (Robust adaptive expected T -count). *For every non-affine $f : \mathbb{F}_2^n \rightarrow \{0, 1\}$ and every $\varepsilon \geq 0$,*

$$\begin{aligned} \mathcal{T}_\varepsilon^{\text{adaptive}}(U_f) &\geq \left(1 - \frac{2\varepsilon}{\delta_f}\right)_+ \nu(U_f), & (x)_+ &= \max\{x, 0\}; \\ \varepsilon \leq \delta_f/4 &\implies \mathcal{T}_\varepsilon^{\text{adaptive}}(U_f) \geq \frac{1}{2}\nu(U_f). \end{aligned}$$

At $\varepsilon = 0$, [Theorem 17](#) recovers the exact nullity lower bound.

Example 18 (AND and Toffoli). For $m \geq 2$ and $f = \text{AND}_m$, the distance from f to the zero function is 2^{-m} . Every nonzero derivative is supported on exactly two points. When $m \geq 3$ such a derivative is non-affine and lies at distance 2^{1-m} from the affine functions; when $m = 2$ all nonzero derivatives are affine and are omitted from the inner minimum in [Definition 14](#). Thus

$$\delta_{\text{AND}_m} = 2^{-m}, \quad \nu(U_{\text{AND}_m}) = m + 1.$$

[Theorem 17](#) gives

$$\mathcal{T}_\varepsilon^{\text{adaptive}}(U_{\text{AND}_m}) \geq \left(1 - 2^{m+1}\varepsilon\right)_+ (m + 1),$$

and hence a lower bound of $(m + 1)/2$ whenever $\varepsilon \leq 2^{-m-2}$. The window remains exponentially small in m . This recovers the $\Omega(\min\{n, \log(1/\varepsilon)\})$ bound of GKZ for Toff_n [\[GKZ25\]](#). Preparing all but m controls in $|1\rangle$, a stabilizer input, turns an adaptive ε -implementation of Toff_n into one of U_{AND_m} with the same T gates and no larger diamond error, and $m = \min\{n - 1, \lfloor \log(1/\varepsilon) \rfloor - 2\}$ satisfies $\varepsilon \leq 2^{-m-2}$, with $m \geq 2$ whenever $n \geq 3$ and $\varepsilon \leq 1/16$.

The same projector argument extends from the largest nontrivial Pauli expectation to the number of large ones. This will give a constant-error bound for majority in [Theorem 33](#).

Lemma 19 (Counting large Pauli expectations). *Let $|J\rangle$ be a pure N -qubit state, let $0 < \tau \leq 1$, and define*

$$K_\tau = |\{Q \in \mathcal{P}_N : |\langle J|Q|J\rangle| > 1 - \tau\}|.$$

Let $|\Psi\rangle$ be a pure state on these N qubits and r additional qubits, put $v = \nu(\Psi)$, and let $F = \langle J| \text{Tr}_{\text{anc}} |\Psi\rangle\langle\Psi| |J\rangle$. Then

$$F \leq 1 - \tau + \tau \min\{1, K_\tau 2^{v-N}\}.$$

Corollary 20 (Expected cost from Pauli counting). *Let U be a q -qubit unitary, put $N = 2q$, fix $0 < \tau \leq 1$, and let K_τ be the quantity in [Lemma 19](#) for $|J_U\rangle$. If a real number $\ell_* > 0$ satisfies $K_\tau 2^{\ell_*-N} \leq 1/2$, then for every $\varepsilon \geq 0$,*

$$\mathcal{T}_\varepsilon^{\text{adaptive}}(U) \geq \left(1 - \frac{2\varepsilon}{\tau}\right)_+ \ell_*.$$

Proof. Run an adaptive implementation on its Choi input and retain discarded qubits as in the proof of [Theorem 17](#). A branch with $\nu(\Psi_b) < \ell_*$ has $F_b \leq 1 - \tau/2$ by [Lemma 19](#). Since $\sum_b p_b(1 - F_b) \leq \varepsilon$, these branches have total probability at most $2\varepsilon/\tau$. Every other branch uses at least $\nu(\Psi_b) \geq \ell_*$ T gates. Averaging and then passing from the maximally mixed input to the worst input gives the claim. $\square$

3.1.4 The exact bracket

Setting $\varepsilon = 0$ in [Theorem 17](#) gives the lower end of a bracket whose upper end is the multiplicative complexity $c_\wedge(f)$, the least number of AND gates in an $\{\text{AND}, \oplus, 1\}$ circuit computing f .

Proposition 21 (Exact T -count bracket). *For every $f : \mathbb{F}_2^n \rightarrow \{0, 1\}$,*

$$\nu(U_f) \leq \mathcal{T}_0^{\text{adaptive}}(U_f) \leq \mathcal{T}_0^{\text{mixed}}(U_f) \leq \mathcal{T}_0^{\text{unitary}}(U_f) \leq 8 c_\wedge(f), \quad \mathcal{T}_0^{\text{adaptive}}(U_f) \leq 4 c_\wedge(f).$$

Proof. The model inequalities are (1), proved after Definition 10. The leftmost inequality is the $\varepsilon = 0$ case of Theorem 17 when f is non-affine. It follows from the monotonicity of nullity under stabilizer operations [BCHK20], and its unitary form is due to Jiang and Wang [JW21]. For affine f it holds trivially, since then $\nu(U_f) = 0$ by Lemma 12. For the upper bounds, realize U_f by a reversible $\{\text{AND}, \oplus\}$ circuit with $c_\wedge(f)$ AND gates. In the adaptive model each AND costs four T gates with measurement-based uncomputation [Jon13, Gid18], the accounting we follow being that of [MSC⁺19], and in the unitary model eight, via a compute and uncompute pair of relative-phase Toffolis [Sel13, Mas16]. $\square$

The bracket closes up to a constant factor for symmetric oracles (Corollary 29) and for inner product (Example 22).

Example 22 (Inner product). Let $f = \text{IP}(x, y) = \bigoplus_{i=1}^n x_i y_i$ on $2n$ input bits. Its derivatives $\partial_{(u,v)} \text{IP} = u \cdot y \oplus v \cdot x \oplus u \cdot v$ are all affine, the first-order ones being $\partial_{x_i} \text{IP} = y_i$ and $\partial_{y_i} \text{IP} = x_i$, but constant only for $u = v = 0$. So $L(\text{IP}) = \{0\}$ and Lemma 12 gives $\nu(U_{\text{IP}}) = 2n + 1$. Against this, $c_\wedge(\text{IP}) = n$ [MS92], so Proposition 21 gives

$$2n + 1 \leq \mathcal{T}_0^{\text{adaptive}}(U_{\text{IP}}) \leq 4n.$$

At $n = 1$ we have $\text{IP}_1 = \text{AND}_2$ and $U_{\text{IP}_1} = \text{CCX}$, so the display reads $3 \leq \mathcal{T}_0^{\text{adaptive}}(\text{CCX}) \leq 4$, consistent with the four- T measured AND gadget used in the proof of Proposition 21.

For quadratic oracles $\delta_f \geq 1/4$, so the window $\varepsilon \leq \delta_f/4$ does not shrink with n .

Corollary 23 (Quadratic oracles at constant error). *Let $f : \mathbb{F}_2^n \rightarrow \{0, 1\}$ have algebraic degree exactly 2, so that f is not affine, and let*

$$B_f(u, v) = f(x) \oplus f(x \oplus u) \oplus f(x \oplus v) \oplus f(x \oplus u \oplus v),$$

which is independent of x , be its associated alternating form, of rank $2h$. Then $\delta_f = \frac{1}{2}(1 - 2^{-h})$ and $\nu(U_f) = 2h + 1$, so

$$\mathcal{T}_\varepsilon^{\text{adaptive}}(U_f) \geq \left(1 - \frac{4\varepsilon}{1 - 2^{-h}}\right)_+ (2h + 1).$$

In particular $\mathcal{T}_\varepsilon^{\text{adaptive}}(U_f) \geq \frac{1}{2}(2h + 1) \geq h$ for every $h \geq 1$ and every $\varepsilon \leq 1/16$.

Proof. Every derivative $\partial_u f(x) = B_f(u, x) \oplus \partial_u f(0)$ is affine in x , so the inner minimum in Definition 14 is $+\infty$ and $\delta_f = \text{dist}(f, \text{Aff})$. Moreover $\partial_u f$ is constant exactly when $B_f(u, \cdot) \equiv 0$, so $L(f) = \text{rad}(B_f)$, which has dimension $n - 2h$, and Lemma 12 gives $\nu(U_f) = n + 1 - (n - 2h) = 2h + 1$. For the distance, a quadratic form of rank $2h$ has $\left|(-1)^f(S)\right| \in \{0, 2^{-h}\}$ for every S , so $\max_\ell |\mathbb{E}(-1)^{f \oplus \ell}| = 2^{-h}$ and $\text{dist}(f, \text{Aff}) = \frac{1}{2}(1 - 2^{-h})$. Substituting into Theorem 17 gives the display. Since $h \geq 1$, we have $\delta_f \geq 1/4$, hence $\delta_f/4 \geq 1/16$. $\square$

For $f = \text{IP}$ on $2n$ bits the form $B_{\text{IP}}((u, v), (u', v')) = u \cdot v' \oplus u' \cdot v$ is nondegenerate of rank $2n$, so Corollary 23 returns $\nu(U_{\text{IP}}) = 2n + 1$ as computed in Example 22, and $\delta_{\text{IP}} = \frac{1}{2}(1 - 2^{-n})$. With the upper bound $\mathcal{T}_0^{\text{adaptive}}(U_{\text{IP}}) \leq 4n$ of Example 22 this gives

$$\mathcal{T}_\varepsilon^{\text{adaptive}}(U_{\text{IP}}) = \Theta(n) \quad \text{for every } \varepsilon \leq \frac{1 - 2^{-n}}{8}.$$

When $h = 1$, an invertible affine input change and addition of an affine function reduce f to AND_2 with unused input variables. These changes preserve the quantities here, and the corollary returns $\delta_f = 1/4$ and $\nu = 3$, matching Example 18. By contrast, AND_m for $m \geq 3$ has degree m , so the corollary says nothing about it. The gap window shrinks as $\delta_f \rightarrow 0$. For majority, where $\delta_f = \Theta(n^{-1/2})$, Theorem 33 uses Lemma 19 instead.

Remark 24 (Comparison with communication-complexity bounds). Girish, May, Parham, and Yuen relate magic to communication complexity [GMPY25], and inner product has two-way communication complexity $\Omega(n)$. Corollary 23 instead counts T gates alone and bounds the worst-input expected T -count directly.

In the mixed model the statement is not new. A parity query on (x, y) costs at most two bits of communication, so for every fixed $\varepsilon_\star < 1/2$ one has $\text{RPDT}_\varepsilon^{\text{na}}(\text{IP}) = \Omega(n)$ uniformly over $0 \leq \varepsilon \leq \varepsilon_\star$, and [GKZ25, Theorem 5] already gives $\mathcal{T}_\varepsilon^{\text{mixed}}(U_{\text{IP}}) = \Omega(n)$ in the same error regime. For majority the communication route is weaker: one party can send its Hamming weight, so the two-party communication complexity is $O(\log n)$ and a communication bound gives at most $O(\log n)$, while Theorem 33 gives $\Theta(n)$.

The mixed-model bound of GKZ transfers to Pauli-frame adaptive circuits without any magic monotone.

Definition 25 (Pauli-frame adaptive circuit). A Pauli-frame adaptive circuit is an adaptive Clifford+ T circuit in the sense of Definition 10 whose locations and total number of T gates are fixed independently of all measurement outcomes and internal classical randomness, whose measurements are computational-basis measurements of single qubits, which may be reused afterwards, and whose only conditioned gates are Pauli operators, each applied when an $\mathbb{F}_2$ -affine function of the measurement record so far equals 1. Classical randomness is otherwise allowed.

Every branch of such a circuit uses the same number t of T gates, so its adaptive cost is t .

Proposition 26 (Pauli-frame adaptive circuits). *If a Pauli-frame adaptive circuit with t T gates implements U_f within diamond error ε , then $t \geq \text{RPDT}_\varepsilon^{\text{na}}(f) - 1$.*

More general corrections obey the same bound with an additive correction term. Suppose the circuit has t unconditional T gates and conditioned gates $G_1, \dots, G_K$, where G_i is applied when an arbitrary Boolean function g_i of the record equals 1, and each G_i is a single T gate or a Pauli or Clifford gate on at most c qubits, fixed as a specific unitary with entries in $\mathbb{Z}[1/\sqrt{2}, i]$. Then

$$t + \kappa_c \sum_{i=1}^K (\mathbf{1}[G_i \text{ is not a Pauli}] + c_\wedge(g_i)) \geq \text{RPDT}_\varepsilon^{\text{na}}(f) - 1$$

for a constant κ_c depending only on c , where $c_\wedge(g)$ vanishes exactly when g is affine.

The standard Clifford and T matrices have entries in $\mathbb{Z}[1/\sqrt{2}, i]$, and the choice of global phase is immaterial, since it does not affect the channel implemented within a classical branch.

3.2 Majority and symmetric Boolean oracles

3.2.1 All symmetric Boolean oracles

Lemma 27 (Exact synthesis of symmetric oracles). *For every symmetric Boolean function $f : \mathbb{F}_2^n \rightarrow \{0, 1\}$,*

$$\mathcal{T}_0^{\text{unitary}}(U_f) = O(n).$$

The construction uses $O(n/\log n)$ clean ancillae in general. If the predicate on the Hamming weight has a Boolean circuit of size $O(\log n)$, as for thresholds and comparisons, then $O(\log n)$ clean ancillae suffice.

Proof. Reversibly compute the Hamming weight $|x|$ into $b = \lceil \log(n+1) \rceil$ qubits using the blocked counting circuit from the proof of Proposition 35, applied to the n input bits. This uses $O(n)$ Toffoli gates and $O(\log n)$ clean workspace. The remaining predicate is an arbitrary Boolean function on b bits. By the Shannon–Lupanov bound [Sav97], every b -bit Boolean function has a fan-in-2 circuit of size $O(2^b/b) = O(n/\log n)$. Compute each gate output into a fresh ancilla, copy the final bit to the oracle target, and uncompute. This reversible simulation uses $O(n/\log n)$ Toffoli gates and ancillae. Reversing the Hamming-weight computation clears the weight register. The total T -count is $O(n)$. If the weight predicate itself has size $O(\log n)$, its reversible evaluation uses only $O(\log n)$ additional ancillae. $\square$

Lemma 28 (Nullity of symmetric oracles). *Let $f : \mathbb{F}_2^n \rightarrow \{0, 1\}$ be symmetric, non-affine, and $n \geq 3$. Then*

$$\dim L(f) \in \{0, 1\}, \quad \nu(U_f) \in \{n, n+1\}.$$

Moreover $\dim L(f) = 1$ exactly when $f(x) \oplus f(x \oplus \mathbf{1})$ is constant.

Corollary 29 (Symmetric oracles at $\varepsilon = 0$). *For every symmetric non-affine f on $n \geq 3$ bits, the three exact T -counts, the stabilizer nullity, and the multiplicative complexity are all $\Theta(n)$:*

$$n \leq \nu(U_f) \leq \mathcal{T}_0^{\text{adaptive}}(U_f) \leq \mathcal{T}_0^{\text{mixed}}(U_f) \leq \mathcal{T}_0^{\text{unitary}}(U_f) \leq 8 c_{\wedge}(f) = O(n).$$

Proof. Lemma 28 gives $\nu(U_f) \geq n$, and the chain in the middle is Proposition 21. The construction of Lemma 27 uses $O(n)$ Toffoli gates for the weight computation and $O(n/\log n)$ for the weight predicate, hence $O(n)$ AND gates in all, so $c_{\wedge}(f) = O(n)$. $\square$

Theorem 30 (Tightness for symmetric oracles in the gap window). *Let f be symmetric and non-affine on $n \geq 3$ bits. Then*

$$\mathcal{T}_{\varepsilon}^{\text{adaptive}}(U_f) \leq \mathcal{T}_{\varepsilon}^{\text{mixed}}(U_f) = O(n)$$

for every ε , and

$$\mathcal{T}_{\varepsilon}^{\text{adaptive}}(U_f) = \Theta(n) \quad \text{and} \quad \mathcal{T}_{\varepsilon}^{\text{mixed}}(U_f) = \Theta(n)$$

whenever $\varepsilon \leq \delta_f/4$.

Proof. The upper bound is exact synthesis from Lemma 27. By Lemma 28, $\nu(U_f) \geq n$. The lower bound follows from Theorem 17, and the mixed lower bound follows because $\mathcal{T}_{\varepsilon}^{\text{mixed}} \geq \mathcal{T}_{\varepsilon}^{\text{adaptive}}$. $\square$

3.2.2 The majority gap and constant-error bounds

For odd n , let

$$\text{MAJ}_n(x) = \mathbf{1}[|x| \geq (n+1)/2].$$

The following anti-concentration statement is proved in Appendix B.

Lemma 31 (Majority derivatives). *There is an absolute constant $c > 0$ such that for every odd n and every nonempty proper $S \subset [n]$,*

$$\min_x \left\{ \Pr_x[\partial_S \text{MAJ}_n(x) = 1], \Pr_x[\partial_S \text{MAJ}_n(x) = 0] \right\} \geq \frac{c}{\sqrt{n}}.$$

Moreover, if $\zeta n \leq |S| \leq (1 - \zeta)n$ for $0 < \zeta \leq 1/2$, then

$$\Pr_x[\partial_S \text{MAJ}_n(x) = 1] \geq c\sqrt{\zeta}.$$

Proposition 32 (Affine gap of majority). *For odd $n \geq 5$,*

$$\delta_{\text{MAJ}_n} = \Theta(n^{-1/2}).$$

Odd majority has the complement direction as its unique nonzero linear structure: $\text{MAJ}_n(x \oplus \mathbf{1}) = 1 \oplus \text{MAJ}_n(x)$. This also follows from the symmetric-function classification in Lemma 28. Thus Lemma 12 gives $\nu(U_{\text{MAJ}_n}) = n$.

With Proposition 32, Theorem 17 gives $\mathcal{T}_{\varepsilon}^{\text{adaptive}}(U_{\text{MAJ}_n}) = \Omega(n)$ only for $\varepsilon = O(n^{-1/2})$. Theorem 33 removes the dependence on n .

Theorem 33 (Adaptive majority at constant error). *Let $c > 0$ be the constant in Lemma 31, and put*

$$\alpha = 1 - h_2(1/4) > 0, \quad \tau = \min\{1/8, c/4\},$$

where $h_2(u) = -u \log u - (1 - u) \log(1 - u)$ is binary entropy, with $0 \log 0 = 0$. For every odd $n \geq 5$ and every $\varepsilon \geq 0$,

$$\mathcal{T}_{\varepsilon}^{\text{adaptive}}(U_{\text{MAJ}_n}) \geq \left(1 - \frac{2\varepsilon}{\tau}\right)_+ (\alpha n - 1)_+. \quad (4)$$

Consequently, there is an absolute, non-explicit $\varepsilon_* > 0$ such that, uniformly for odd $n \geq 5$ and $0 \leq \varepsilon \leq \varepsilon_*$,

$$\mathcal{T}_{\varepsilon}^{\text{adaptive}}(U_{\text{MAJ}_n}) = \mathcal{T}_{\varepsilon}^{\text{mixed}}(U_{\text{MAJ}_n}) = \mathcal{T}_{\varepsilon}^{\text{unitary}}(U_{\text{MAJ}_n}) = \Theta(n).$$

In the mixed model GKZ list $\mathcal{T}_\varepsilon^{\text{mixed}}(U_{\text{MAJ}_n}) = \Omega(n)$ at constant error [GKZ25, Table 1], and Theorem 33 extends the bound to adaptive circuits.

4 Workspace for sampled circuits

Proofs of the longer results are in Appendix C.

4.1 Space-efficient evaluation of the GKZ top function

4.1.1 Fourier sampling produces a signed threshold

We first restate the part of the GKZ construction needed below.

Theorem 34 (GKZ Fourier sampler, restated [GKZ25]). *Let $f : \mathbb{F}_2^n \rightarrow \{0, 1\}$ be nonzero and let $0 < \varepsilon < 1$. Put $L = \|\widehat{f}\|_1$ and set*

$$k = \left\lceil 8L^2 \ln \frac{8}{\varepsilon} \right\rceil.$$

Independently sample $S_1, \dots, S_k$ from

$$\Pr[S_j = S] = \frac{|\widehat{f}(S)|}{L}, \quad \sigma_j = \text{sign}(\widehat{f}(S_j)).$$

For a fixed sample, define

$$g(x) = \mathbf{1} \left[\frac{L}{k} \sum_{j=1}^k \sigma_j \chi_{S_j}(x) \geq \frac{1}{2} \right]. \quad (5)$$

The pointwise error satisfies

$$\max_x \Pr[g(x) \neq f(x)] \leq \frac{\varepsilon}{4}. \quad (6)$$

Consequently the mixed channel $\mathcal{E}(\rho) = \mathbb{E}[U_g \rho U_g^\dagger]$ satisfies $D_\diamond(\mathcal{E}, U_f) \leq \varepsilon$.

Lemma 11 uses this per-input guarantee directly, because a union bound over all 2^n inputs would need $k = \Theta(L^2 n)$ and lose the independence from n .

For a fixed sample, let

$$z_j = \langle S_j, x \rangle, \quad c_j = \mathbf{1}[\sigma_j = -1], \quad b_j = z_j \oplus c_j.$$

Then $\sigma_j(-1)^{z_j} = (-1)^{b_j} = 1 - 2b_j$. Writing $P = \sum_j b_j$, the branch condition (5) is

$$P \leq \theta, \quad \theta = \left\lfloor \frac{k}{2} - \frac{k}{4L} \right\rfloor. \quad (7)$$

Since f is nonzero, evaluating its Fourier expansion at an input with $f(x) = 1$ gives $L \geq 1$. Consequently $k/4 - 1 \leq \theta < k/2$, so both sides of the comparison have thresholds at linear distance from the endpoints of the Hamming-weight range.

4.1.2 A blocked reversible tally

Proposition 35 (Workspace-efficient tally). *Under the assumptions of Theorem 34, there is a mixed Clifford+T implementation of U_f at diamond error ε with*

$$T = O(k) = O\left(L^2 \log(8/\varepsilon)\right), \quad \lambda = O(\log k).$$

Incrementing a binary accumulator once for each b_j would lay down a worst-case carry chain of length $\Theta(\log k)$ at every step, for $\Theta(k \log k)$ non-Clifford gates in total, which is why the tally adds whole blocks.

The threshold has a classical circuit with linearly many AND gates [BPP00], the source of GKZ's $O(k)$ count. Bravyi, Yoder and Maslov record logarithmic-ancilla reversible weight computation of m bits at $O(m \log^2 m)$ gates, and their own construction reaches $O(m \log m)$ [BYM22]. Zi, Yuan, Nie and Zhang trade depth against ancillae for Hamming weight [ZYNZ26].

4.1.3 One clean ancilla with a read-only input

The tally rereads its input to clear each block and to reverse the accumulation. With unrestricted rereading, constant workspace is already available. Over $\mathbb{F}_2$, the three-register programs of Ben-Or and Cleve [BOC92] map (r_1, r_2, r_3) to $(r_1 \oplus g(x)r_2, r_2, r_3)$ for every register content, using CNOTs and Toffoli gates controlled by one input bit and one register, with length $4^{O(d)}$ for a formula of depth d . With r_2 a clean ancilla prepared in $|1\rangle$, r_1 the target and r_3 borrowed, this computes U_g exactly at polynomial T -count. Running the program twice, with an X on r_2 before the second run and after it, makes r_2 borrowable as well. Proposition 36 removes one register: a linear representation of A_5 puts the computation on the target and one further qubit, approximately, and Remark 37 lets that qubit be borrowed.

Proposition 36 (Coherent Barrington evaluation). *For every nonzero Boolean $f : \mathbb{F}_2^n \rightarrow \{0, 1\}$ and $0 < \varepsilon < 1$, there is a mixed Clifford+T implementation of U_f with diamond error ε that*

- *uses at most one clean ancilla qubit,*
- *never writes to the input register, and*
- *uses $\text{poly}(n, k)$ T gates per sampled circuit, where $k = \Theta(\|\hat{f}\|_1^2 \log(8/\varepsilon))$.*

For $f \equiv 0$, the identity circuit gives an exact implementation with no ancilla and no T gates.

Remark 37 (Borrowed second qubit). The exact program acts on the target and the second qubit as the operator I or $X \otimes I$, whatever state the second qubit is in, and each compiled instruction is within its operator-norm budget of the exact one. The second qubit may therefore be borrowed from outside the input register, in an arbitrary state possibly entangled with the rest of the computation. Each branch then acts on the input, the target and the borrowed qubit within operator-norm distance $\varepsilon/2$ of $U_g \otimes I$, so the averaged channel is within diamond distance ε of $U_f \otimes I$. The borrowed qubit is restored only approximately, unlike the exact restoration required of data qubits in Section 2. With that relaxation the construction of Proposition 36 uses no clean ancilla.

Remark 38 (Projective representations). Width-2 quantum branching programs accept every NC^1 language with zero error through the icosahedral action of A_5 on a qubit [AGK⁺05]. That action is projective: it is a linear representation of the binary icosahedral double cover, not of A_5 itself. Projective phases are invisible to acceptance probabilities, but an input-dependent sign changes relative phases between computational-basis inputs and therefore changes the oracle. The width-2 acceptance result therefore gives no one-qubit coherent synthesis.

Proposition 39 (Exact synthesis with only a target workspace). *Consider circuits on an input register $x \in \mathbb{F}_2^n$ and one target qubit, with no other qubits, whose elementary gates are one-qubit Clifford+T gates on the target and CNOTs from individual input qubits to the target. In this gate model, a circuit implements the Boolean oracle U_g exactly, up to one input-independent global phase, if and only if g is affine.*

The proposition concerns exact synthesis in this elementary gate model. Proposition 36 gives an approximate construction with one additional workspace qubit, clean or, by Remark 37, borrowed, so it does not show that approximation alone removes the obstruction.

4.2 One-ancilla synthesis from affine-subspace tests

4.2.1 Randomized Toffoli from a linear-subspace test

For the Toffoli gate, the dedicated GKZ sampler has an OR top function. Its zero set is a linear subspace, so a change of coordinates reduces each branch to one multi-controlled gate.

Construction 40 (Sampled OR branch). Put $\ell = \lceil \log(4/\varepsilon) \rceil$ and assume $\ell \leq n - 1$; the complementary case is handled at the end of the proof of [Proposition 41](#). Sample an ordered ℓ -tuple $a_1, \dots, a_\ell \in \mathbb{F}_2^{n-1}$ uniformly conditioned on linear independence. This conditioning has constant expected resampling overhead because

$$\Pr[\text{independent}] = \prod_{i=0}^{\ell-1} (1 - 2^{i-n+1}) \geq \prod_{j=1}^{\infty} (1 - 2^{-j}) > 0.28.$$

First apply X to all $n - 1$ control qubits, writing $y = x \oplus \mathbf{1}$. Complete $a_1, \dots, a_\ell$ to a basis of $\mathbb{F}_2^{n-1}$ and implement the corresponding invertible linear transformation by a CNOT circuit. The sampled parities $a_j \cdot y$ become the first ℓ data coordinates. A single ℓ -controlled NOT with negative controls flips the target exactly when all sampled parities vanish. Equivalently, it computes the Boolean function $1 \oplus g(y)$ for the GKZ function $g = \text{OR}_j(a_j \cdot y)$, and hence realizes the sampled GKZ branch. Undo the basis change and the initial X layer.

Proposition 41 (Randomized Toffoli with one clean ancilla). *For every $n \geq 2$ and $0 < \varepsilon < 1$, there is a mixed Clifford+T implementation of Toff_n with at most one clean ancilla and*

$$T = O\left(\min\left\{n, \log \frac{2}{\varepsilon}\right\}\right)$$

T gates per branch. Every branch implements a Boolean oracle exactly and restores all work qubits exactly. If $\ell = \lceil \log(4/\varepsilon) \rceil \leq n - 2$, no clean ancilla is needed.

The pointwise-to-diamond conversion is not specific to OR: [Lemma 11](#) applies to every Boolean-oracle mixture, and [Corollary 52](#) combines it with (6) for a general sampler conditioned on linear independence.

4.2.2 Bounded spectral norm and a reusable clean ancilla

The preceding construction extends beyond a single sampled top function, through the quantitative idempotent theorem. For $M \geq 1$ let $q(M)$ be the least integer J such that every integer-valued $f : \mathbb{F}_2^n \rightarrow \mathbb{Z}$, for every n , with $\|\widehat{f}\|_1 \leq M$ can be written with at most J terms as

$$f = \sum_{j=1}^J \sigma_j \mathbf{1}_{a_j + V_j}, \quad \sigma_j \in \{-1, 1\}, \quad (8)$$

with $a_j + V_j$ affine subspaces of $\mathbb{F}_2^n$.

Theorem 42 (Green–Sanders [[GS08](#)]; Sanders [[San20](#), [San19](#)]). *There is an absolute constant C such that $q(M) \leq \lceil \exp(\exp(CM^4)) \rceil$ for every $M \geq 1$ [[GS08](#), Theorem 1.3]. Moreover $q(M) \leq \exp(M^{4+o(1)})$ [[San20](#), Theorem 1.1] and $q(M) \leq \exp(M^{3+o(1)})$ [[San19](#)] as $M \rightarrow \infty$.*

The Green–Sanders bound and the $\exp(M^{4+o(1)})$ bound hold in every finite abelian group, and the $\exp(M^{3+o(1)})$ bound is stated for integer-valued functions on $\mathbb{F}_2^n$, as a signed sum of subspace indicators. In each, the algebra norm is $\|\widehat{f}\|_1$ in the normalization of [Section 2](#). A coset of a subgroup of $\mathbb{F}_2^n$ is an affine subspace.

Proposition 43 (Bounded-spectral-norm synthesis). *For every $M \geq 1$, every Boolean $f : \mathbb{F}_2^n \rightarrow \{0, 1\}$ with $\|\widehat{f}\|_1 \leq M$, and every $0 < \varepsilon < 1$, there is a mixed Clifford+T implementation of U_f with at most one clean ancilla and*

$$T = O\left(q(M) \log \frac{2q(M)}{\varepsilon}\right)$$

T gates per branch. Each branch restores the clean ancilla exactly. In particular, for fixed M the bound is $O_M(\log(2/\varepsilon))$, uniformly in the input length n.

By [Theorem 42](#) the T-count is $O(q(M) \log(2q(M)/\varepsilon))$ with $q(M) \leq \exp(M^{3+o(1)})$, far above the $O(M^2 \log(8/\varepsilon))$ of Fourier sampling, and [Open Problem 7](#) asks whether the dependence on M can be made polynomial.

4.3 Memory lower bounds for threshold evaluators

This subsection bounds the memory of abstract evaluators of a k -bit Hamming threshold, the top function of a sampled branch. The measure is the evaluator's total memory w , borrowed or allocated, and correctness is required on all 2^k bit patterns. [Section 4.3.3](#) gives the condition under which the bounds apply to an actual branch.

4.3.1 Exact single-pass evaluation

Definition 44 (Single-pass evaluator). Fix an order of k input positions and relabel them in that order. A single-pass evaluator consists of workspace registers $W_0, \dots, W_k$, a fixed input-independent initial state on W_0 , quantum channels

$$\mathcal{V}_i : W_{i-1} \otimes \mathbb{C}_i^2 \longrightarrow W_i \otimes \mathbb{C}_i^2,$$

and a final two-outcome measurement of W_k . Each channel acts only on the workspace and the current position; position i is never touched again, so a channel may alter it freely, since nothing later depends on what it leaves behind. Fresh environments may be used to implement a channel and then discarded, but no environment, measurement record, or other memory is retained outside W_i . In particular, subsequent operations cannot depend on a discarded record. The initial workspace is uncorrelated with the input. The evaluator computes $h : \{0, 1\}^k \rightarrow \{0, 1\}$ exactly if the final outcome equals $h(v)$ with certainty on every computational-basis input v . Its workspace is $w = \max_i [\log \dim W_i]$ qubits. This model includes classical irreversible streaming.

For an integer $1 \leq \theta \leq k$, define

$$\text{THR}_\theta(v) = \mathbf{1} \left[\sum_i v_i \geq \theta \right].$$

Proposition 45 (Sharp single-pass threshold bound). *Every exact single-pass evaluator for THR_θ uses*

$$w \geq \lceil \log(\min\{\theta + 1, k - \theta + 2\}) \rceil$$

qubits of workspace.

A branch accepts when $P \leq \theta$, by (7), which is the negation of $P \geq \theta + 1$; so negating the output and applying [Proposition 45](#) to $\text{THR}_{\theta+1}$ bounds the workspace of any machine that reads the k bits once each in a fixed order and answers correctly on all 2^k patterns, the *full cube*. Both θ and $k - \theta$ grow linearly in k by [Section 4.1.1](#), so that bound is $\Omega(\log k)$.

4.3.2 Single-pass evaluation at bounded error

The full-cube single-pass bound does not need exactness. It survives every fixed classical output error below $1/2$, by a serial-encoding argument, and a sampled GKZ branch need not evaluate its top function exactly. [Remark 49](#) relates this error model to the diamond error of a branch.

Definition 46 (η -error evaluator). An η -error single-pass evaluator is a single-pass evaluator in the sense of [Definition 44](#) whose final measurement returns $h(v)$ with probability at least $1 - \eta$ on every computational-basis input $v \in \{0, 1\}^k$. The case $\eta = 0$ is the exact model.

Lemma 47 (Serial threshold encoding [[Nay99](#), [ANTSV02](#)]). Let $M \geq 1$, and let $\rho_0, \dots, \rho_{M-1}$ be states on a D -dimensional register W . Suppose that, for every $t \in \{1, \dots, M-1\}$, there is a measurement effect E_t such that

$$|\text{Tr}(E_t \rho_s) - \mathbf{1}[s \geq t]| \leq \eta \quad (0 \leq s < M),$$

where $0 \leq \eta < 1/2$. Then

$$\log D \geq (1 - h_2(\eta)) \lfloor \log M \rfloor,$$

where h_2 is the binary entropy function of [Theorem 33](#).

Theorem 48 (Single-pass bound with branch error). Let $1 \leq \theta \leq k$, put $m = \min\{\theta, k + 1 - \theta\}$, and let $0 \leq \eta < 1/2$. Every η -error single-pass evaluator for THR_θ uses workspace

$$w \geq (1 - h_2(\eta)) \lfloor \log(m + 1) \rfloor.$$

Consequently, when $m = \Theta(k)$, every fixed $\eta < 1/2$ requires $w = \Omega(\log k)$.

A one-pass counter modulo $k + 1$ attains this scale.

Remark 49 (Branch error and diamond error). A branchwise coherent approximation implies the classical guarantee used by [Theorem 48](#), but the direction of implication matters. Suppose that, for every sampled branch g , the implemented branch channel $\tilde{\mathcal{U}}_g$ and the ideal channel $\mathcal{U}_g = \Phi_{U_g}$ satisfy

$$D_\diamond(\tilde{\mathcal{U}}_g, \mathcal{U}_g) \leq \eta.$$

On a computational-basis input with target initialized to $|0\rangle$, its output bit is wrong with probability at most η . Thus [Theorem 48](#) applies when the branch computation has the specified evaluator form. Convexity also gives

$$D_\diamond(\mathbb{E}_g \tilde{\mathcal{U}}_g, \mathbb{E}_g \mathcal{U}_g) \leq \eta.$$

Hence taking $\eta = O(\varepsilon)$ adds $O(\varepsilon)$ to the sampler error. The single-pass lower bound only needs η to be bounded away from $1/2$; no relationship of order $1/k$ is needed.

The converse fails, since small classical error on basis inputs does not control the phases of a coherent branch. Accuracy of the averaged channel also does not force each branch to be accurate, because branch errors may cancel in the average, so the bound concerns branchwise approximation.

4.3.3 Transfer to sampled branches

A sampled GKZ branch need only be correct on the code

$$\{(\langle S_1, x \rangle, \dots, \langle S_k, x \rangle) : x \in \mathbb{F}_2^n\} \subseteq \mathbb{F}_2^k,$$

not necessarily on the full cube. The next condition lets us transfer [Proposition 45](#) and [Theorem 48](#) to an evaluator whose only access to the input is through these parity positions.

Definition 50 (Spread Fourier sampler). Let f be nonzero and $D_f(S) = |\widehat{f}(S)| / \|\widehat{f}\|_1$. For $1 \leq k \leq n$ and $0 < \gamma < 1$, say that D_f is (k, γ) -spread if every subspace $V \leq \mathbb{F}_2^n$ of dimension at most $k - 1$ satisfies

$$\Pr_{S \sim D_f} [S \in V] \leq \frac{\gamma}{k}.$$

Lemma 51 (Conditioning on linear independence). *If D_f is (k, γ) -spread, then k independent samples from D_f are linearly independent with probability at least $1 - \gamma$. If a mixed channel $\mathcal{E}$ is formed from the independent samples and $\mathcal{E}'$ is formed by conditioning on independence, then*

$$D_\diamond(\mathcal{E}, \mathcal{E}') \leq \gamma.$$

Corollary 52 (Lower bound for sampled instances). *Run the GKZ sampler of Theorem 34 with error parameter ε , and suppose $k \leq n$ and D_f is (k, γ) -spread in the sense of Definition 50. After conditioning on linear independence, the total diamond error is at most $\varepsilon/(1 - \gamma)$. Every exact single-pass evaluator for the sampled top function uses $\Omega(\log k)$ workspace; the same conclusion holds with any fixed branch error $\eta < 1/2$. Here the evaluator accesses the input only through the sampled parity positions.*

Example 53 (Two disjoint ANDs). Let $n = 2\ell$ with $\ell \geq 2$, and define

$$f(x) = \text{AND}_\ell(x_1, \dots, x_\ell) \oplus \text{AND}_\ell(x_{\ell+1}, \dots, x_{2\ell}).$$

Put $A = 1 - 2\text{AND}_\ell$, $H = 1 - 2f$, and $u = 2^{-\ell}$. Every Fourier coefficient of AND_ℓ equals $(-1)^{|S|}u$, so $\widehat{A}(\emptyset) = 1 - 2u$ and $|\widehat{A}(S)| = 2u$ for $S \neq \emptyset$, giving $\|\widehat{A}\|_1 = 3 - 4u$. Since H is the product of two functions on disjoint blocks, $\|\widehat{H}\|_1 = (3 - 4u)^2$ and $\widehat{H}(\emptyset) = (1 - 2u)^2$. With $\widehat{f}(\emptyset) = (1 - \widehat{H}(\emptyset))/2$ and $\widehat{f}(S) = -\widehat{H}(S)/2$ elsewhere,

$$\|\widehat{f}\|_1 = (2u - 2u^2) + \frac{1}{2}[(3 - 4u)^2 - (1 - 2u)^2] = 4(1 - 2^{-\ell})^2 < 4.$$

Hence $k = \Theta(\log(8/\varepsilon))$ for its GKZ sampler.

The spread condition is nonvacuous here. Its Fourier coefficients have magnitudes $2u(1 - u)$ at the empty set, $u(1 - 2u)$ when a nonempty set lies in one block, and $2u^2$ when it meets both blocks. Dividing by $4(1 - u)^2$ gives $D_f(S) \leq u$ for every S . Consequently every d -dimensional subspace has mass at most $2^{d-\ell}$, and the explicit sufficient condition

$$k 2^{k-1-\ell} \leq \gamma$$

implies (k, γ) -spreadness. A fixed $\gamma \in (0, 1)$ suffices: Corollary 52 then increases the sampler error by only a constant factor. With $k = \Theta(\log(8/\varepsilon))$, this condition holds for sufficiently large n throughout $\varepsilon \geq 2^{-\kappa n}$ for a sufficiently small absolute $\kappa > 0$.

The example shows that the spread condition can hold, and it is an easy case for oracle synthesis. Its structure already gives a small-workspace circuit: compose the two multi-controlled Toffoli oracles acting on the common target, applying Proposition 41 to each with error $\varepsilon/2$. The total error is at most ε , the total T -count is $O(\min\{n, \log(2/\varepsilon)\})$, and one clean ancilla can be reused because every branch returns it exactly.

The spread hypothesis is restrictive. The subspace $\{0\}$ has mass $\widehat{f}(\emptyset)/\|\widehat{f}\|_1 = \mathbb{E}[f]/\|\widehat{f}\|_1$, so (k, γ) -spreadness forces $\mathbb{E}[f] \leq \gamma \|\widehat{f}\|_1 / k$. With $k = \Theta(\|\widehat{f}\|_1^2 \log(8/\varepsilon))$ this is $O(1/(\|\widehat{f}\|_1 \log(1/\varepsilon)))$, which excludes every balanced function. Corollary 52 therefore applies only to sparse functions such as Example 53, and for that example a direct construction already uses one clean ancilla.

4.3.4 Clean versus borrowed workspace

Remark 54 (Readout from a borrowed register). Let W be a finite-dimensional register whose initial state is arbitrary and unknown. On input v , suppose an evaluator applies a unitary U_v to W and then flips a clean target according to a fixed projective measurement $\{\Pi, I - \Pi\}$ of W . If the evaluator is correct for every initial state of W , then the computed Boolean function is constant. Indeed, an accepted input v requires $U_v|y\rangle \in \text{range}(\Pi)$ for every $|y\rangle \in W$, and since U_v is surjective, $\Pi = I$. A rejected input likewise forces $\Pi = 0$. The same holds if the flip is implemented coherently by $\Pi \otimes X + (I - \Pi) \otimes I$.

We next consider counters whose final states lie on the orbit of a single fixed unitary. This is a stronger requirement than merely depending on the input through a count: an arbitrary encoding $|\phi_{g(v)}\rangle$ need not have the form $V^{g(v)}|\psi\rangle$.

Definition 55 (Count-oblivious evaluator). Let $g : \{0, 1\}^k \rightarrow \mathbb{Z}_{\geq 0}$ have range exactly $\{0, 1, \dots, N\}$. An evaluator is count-oblivious for g if there are a fixed unitary V on a workspace of dimension D , a fixed initial state $|\psi\rangle$, and a fixed effect $0 \leq E \leq I$ such that on every input v the final workspace state is $V^{g(v)}|\psi\rangle$, the *orbit form*, and the evaluator accepts with probability $\langle \psi | V^{-g(v)} E V^{g(v)} | \psi \rangle$. We call it weight-oblivious when $g(v) = |v|$, so $N = k$.

For one full pass, an evaluator that applies a fixed V_1 for every 1 it reads and a fixed commuting V_0 for every 0 is weight-oblivious with $V' = V_0^{-1}V_1$ and $|\psi'\rangle = V_0^k|\psi\rangle$. For p identical passes the final state is

$$V_0^{pk} (V_0^{-1}V_1)^{p|v|} |\psi\rangle,$$

so the same definition applies with $V' = (V_0^{-1}V_1)^p$ and $|\psi'\rangle = V_0^{pk}|\psi\rangle$. More generally, if pass r uses a pair $(V_{r,1}, V_{r,0})$ and all these operations lie in one commuting family, the passes fold into $V' = \prod_r V_{r,0}^{-1}V_{r,1}$ and $|\psi'\rangle = (\prod_r V_{r,0}^k)|\psi\rangle$.

Proposition 56 (Dimension bounds for a fixed-unitary counter). *Let $1 \leq \theta \leq N$ and $m = \min\{\theta, N + 1 - \theta\}$.*

1. *Every count-oblivious evaluator computing $\mathbf{1}[g(v) \geq \theta]$ exactly has $D \geq N + 1$. The cyclic shift on $N + 1$ states, initialized at $|0\rangle$ and read out by the projector onto $\{|\theta\rangle, \dots, |N\rangle\}$, attains this bound.*
2. *If the evaluator has error at most η on every input, where $0 \leq \eta < 1/2$, then*

$$D \geq \frac{N + 1}{1 + 4N\eta(1 - \eta)}, \quad \log D \geq (1 - h_2(\eta)) \lfloor \log(m + 1) \rfloor.$$

For Hamming-weight counters, exact orbit evaluation therefore needs $k + 1$ dimensions, and a rotation or phase encoding of the same orbit cannot reduce this requirement. At $\eta = O(1/k)$ the Gram argument still gives a linear dimension bound, and at any fixed $\eta < 1/2$ the serial-encoding argument gives logarithmic workspace for nondegenerate thresholds.

We can also allow different commuting operations at different positions. Say an evaluator is *abelian* if there are pairwise commuting unitaries $C_1, \dots, C_k$ on a workspace of dimension D , a fixed initial state $|\psi\rangle$, and a fixed effect E , such that its acceptance probability on v is $\langle \psi | \Phi(v)^\dagger E \Phi(v) | \psi \rangle$, where $\Phi(v) = \prod_j C_j^{v_j}$. An implementation with a fixed, input-independent read schedule has this form whenever all per-read operations and all fixed interleaved workspace operations belong to one commuting family: combine the operations controlled by each position and absorb the input-independent product into the initial state. This reduction allows any number of repeated reads. If $g(v) = c + \sum_j w_j v_j$, the count-oblivious orbit is an instance of this form with $C_j = V^{w_j}$ and initial state $V^c|\psi\rangle$.

Proposition 57 (Position-dependent abelian evaluators). *Let $1 \leq \theta \leq k$ and $m = \min\{\theta, k + 1 - \theta\}$. Every abelian evaluator computing THR_θ exactly has $D \geq k + 1$, and this is attained by a cyclic counter. More generally, error at most η on every input, where $0 \leq \eta < 1/2$, implies*

$$D \geq \frac{k + 1}{1 + 4k\eta(1 - \eta)}, \quad \log D \geq (1 - h_2(\eta)) \lfloor \log(m + 1) \rfloor.$$

These bounds hold at every length. In particular, $\eta = O(1/k)$ requires $D = \Omega(k)$ for every nontrivial threshold, and fixed $\eta < 1/2$ requires $\Omega(\log k)$ workspace qubits when $m = \Omega(k)$.

For unweighted thresholds the tight exact dimension bound thus holds throughout the abelian model, which contains the powers of one fixed unitary as a special case. Unitarity is necessary here, since the saturated counter in the proof of [Proposition 45](#) applies either the identity or one fixed channel at each read, so its updates commute, and it decides THR_θ in dimension $\theta + 1$. The construction of [Proposition 36](#) uses noncommuting operations and need not have the fixed-unitary count-orbit form.

A related qualitative obstruction appears for measure-once quantum finite automata: with an isolated cut point, a single fixed automaton over inputs of arbitrary length recognizes exactly the group languages [[BP99](#)]. The language of strings containing at least a fixed positive number of ones is not a group language.

4.3.5 A pass–workspace hierarchy

[Definition 44](#) reads each position once and permits general channel updates. The bounds below allow repeated reads within a read-only isometric model, which we now specify. The workspace is a register disjoint from the k positions; this is what makes “send the workspace across the cut” a well-defined message of w qubits in [Theorem 62](#). A tally held in qubits borrowed from the k positions themselves is therefore outside the model.

Definition 58 (Multi-pass evaluator). Fix a schedule $j_1, \dots, j_R \in [k]$, possibly with repetitions. A *multi-pass evaluator* consists of workspace registers $W_0, \dots, W_R$, a fixed initial state on W_0 , and isometries

$$V_i : W_{i-1} \otimes \mathbb{C}_{j_i}^2 \longrightarrow W_i \otimes \mathbb{C}_{j_i}^2$$

of the read-only controlled form

$$V_i = \sum_{b \in \{0,1\}} V_{i,b} \otimes |b\rangle\langle b|_{j_i}, \quad V_{i,b} : W_{i-1} \longrightarrow W_i.$$

Thus a computational-basis value at position j_i may control the workspace evolution but is never changed. Repeated reads require the input to be preserved. [Definition 44](#) does not require it, because each position is read only once. The evaluator ends with a two-outcome measurement of W_R . Its workspace is $w = \max_i \lceil \log \dim W_i \rceil$ qubits. It computes h exactly if the outcome equals $h(v)$ with certainty on every v , and with error η if the outcome equals $h(v)$ with probability at least $1 - \eta$ on every v . A p -pass evaluator is one whose schedule is p consecutive scans in a common fixed permutation of $[k]$. Relabeling positions in that permutation makes each scan $1, \dots, k$.

Definition 59 (Crossing number). Fix a schedule that reads the k positions, possibly with repetitions, and a balanced cut of the positions. After relabeling positions, write this cut as

$$A = \{1, \dots, \lfloor k/2 \rfloor\}, \quad B = \{\lfloor k/2 \rfloor + 1, \dots, k\}.$$

The crossing number c is the number of times consecutive reads lie on opposite sides of this cut. For a p -pass evaluator, choose A to be the first $\lfloor k/2 \rfloor$ positions in its common scan order and B the remaining positions. This choice gives $c \leq 2p - 1$. Hamming thresholds are unchanged by this relabeling.

We use the standard rank bound for quantum communication.

Lemma 60 (Rank bound [[Kre95](#), [BdW01](#)]). *In a two-party quantum protocol without prior entanglement or shared randomness that communicates q qubits in total, the acceptance-probability matrix has rank at most 4^q . Hence an exact protocol for a Boolean matrix M satisfies*

$$q \geq \frac{1}{2} \log \text{rank}(M).$$

Private randomness is allowed in [Lemma 60](#), since each party can purify it into local workspace.

Lemma 61 (Threshold matrix rank). *Let $1 \leq \theta \leq k$, $k_1 = \lfloor k/2 \rfloor$, $k_2 = \lceil k/2 \rceil$, and*

$$M_\theta[x, y] = \mathbf{1}[|x| + |y| \geq \theta], \quad x \in \{0, 1\}^{k_1}, y \in \{0, 1\}^{k_2}.$$

Then

$$\text{rank}(M_\theta) \geq \min\{\theta, k + 1 - \theta\}.$$

In particular, if $k/4 - 1 \leq \theta \leq 3k/4$, then $\text{rank}(M_\theta) \geq k/4 - 1 = \Omega(k)$.

Proof. Compress rows and columns by Hamming weight. Let $N[a, b] = \mathbf{1}[a + b \geq \theta]$ for $0 \leq a \leq k_1$, $0 \leq b \leq k_2$. The indicator matrices that expand weight classes have full column rank, so $\text{rank}(M_\theta) = \text{rank}(N)$. Put $m = \min\{\theta, k + 1 - \theta\}$ and $a_0 = \max\{0, \theta - k_2\}$. Choose rows of weights $a_0, \dots, a_0 + m - 1$ and columns of weights $\theta - a''$ for a'' in the same range. These weights exist: if $\theta \leq k_2$, then $a_0 = 0$, $m = \theta$, and $a_0 + m - 1 = \theta - 1 \leq k_1$; if $\theta > k_2$, then $a_0 + m - 1 = k_1 < \theta$. In both cases the column weights lie in $[0, k_2]$. The resulting submatrix has entries $\mathbf{1}[a \geq a'']$ and is triangular with diagonal entries one. Its rank is m . $\square$

Theorem 62 (Pass–workspace tradeoff). *Let $k \geq 16$ and let an exact evaluator for THR_θ , in the sense of Definition 58, with $k/4 - 1 \leq \theta \leq 3k/4$, use w workspace qubits and have crossing number c . Then*

$$wc \geq \frac{1}{2} \log(k/4 - 1) = \Omega(\log k).$$

Consequently every exact p -pass evaluator satisfies $wp = \Omega(\log k)$.

The bound is tight up to a constant factor when $c = O(1)$: a one-pass counter modulo $k + 1$ has $c = 1$ and $w = \lceil \log(k + 1) \rceil$.

The rank argument also tolerates small branch error, though unlike the single-pass serial-encoding bound of Theorem 48 it retains an error threshold of order $1/k$. It compares the acceptance matrix against an ideal one, so we first record when a perturbation can lower the rank. Write Λ_N for the $N \times N$ upper-triangular all-ones matrix, $\Lambda_N[i, j] = \mathbf{1}[j \geq i]$; this is the acceptance pattern of a threshold, accept exactly when $j \geq i$.

Lemma 63 (Perturbed staircase). $\sigma_{\min}(\Lambda_N) \geq 1/2$. *Consequently every real $N \times N$ matrix A with $\|A - \Lambda_N\|_F < 1/2$ has rank N ; in particular this holds whenever every entry of $A - \Lambda_N$ has magnitude less than $1/(2N)$.*

Remark 64 (Sharpness of the perturbation scale). The entrywise scale in Lemma 63 is sharp up to a factor of two. Let $x \in \mathbb{R}^N$ have alternating entries $x_j = (-1)^j$, put $y = \Lambda_N x$, and define

$$E = -\frac{1}{N} y x^\top.$$

Every suffix sum in y has magnitude at most 1, so $\|E\|_{\max} \leq 1/N$; moreover $x^\top x = N$, and hence

$$(\Lambda_N + E)x = y - y \frac{x^\top x}{N} = 0.$$

Thus an entrywise perturbation of size $1/N$ can already drop the rank, so this exact-full-rank perturbation estimate only reaches errors of order $1/N$. A drop from N to $N - 1$ does not preclude other rank lower bounds. More directly, Lemma 47 and Theorem 48 use the state–measurement factorization to establish the constant-error single-pass bound without requiring full rank.

Corollary 65 (Pass–workspace tradeoff with branch error). *Let $k \geq 16$, $k/4 - 1 \leq \theta \leq 3k/4$, and put $m = \min\{\theta, k + 1 - \theta\} = \Omega(k)$. Every η -error evaluator for THR_θ with $\eta < 1/(2m)$, workspace w , and crossing number c satisfies*

$$wc \geq \frac{1}{2} \log m,$$

so every η -error p -pass evaluator with $\eta < 1/(2m)$ still obeys $wp = \Omega(\log k)$.

The theorem controls only the product of workspace and crossings. The threshold matrix has an $O(\log k)$ deterministic communication protocol in which one party sends its Hamming weight, so a lower bound on total communication alone cannot force growing workspace once the number of crossings exceeds this scale. At constant error the limit is lower still. After computing their Hamming weights locally, the parties need only decide Greater-Than on $O(\log k)$ -bit integers. That problem has a bounded-error protocol exchanging $O(\log \log k)$ bits [Nis93], and private randomness suffices by Newman's theorem [New91] applied to these $O(\log k)$ -bit inputs. At constant error a communication argument therefore cannot prove a lower bound beyond $wc = \Omega(\log \log k)$, and a bound of the form $wp = \Omega(\log k)$ would have to use more of the evaluator's structure than its communication.

Corollary 65 holds only at branch error $O(1/k)$, which leaves the following question.

Open Problem 66 (The pass–workspace tradeoff at constant error). In the model of Definition 58, does $wp = \Omega(\log k)$ persist for a central threshold on k bits at every fixed branch error $\eta < 1/2$?

A Proofs of the lower bounds

A.1 Stabilizer nullity of Boolean oracles

Proof of Lemma 12. Let t denote the target qubit. A general phase-free Pauli has the form $P = X_S Z_{S'} \otimes X_t^b Z_t^a$. On a computational-basis state, conjugation by U_f changes the target displacement by $\partial_S f(x)$ and contributes the phase $(-1)^{af(x)}$, up to the affine phase already present in $Z_{S'} Z_t^a$. Consequently $U_f P U_f^\dagger$ is a Pauli exactly when

1. $\partial_S f$ is constant, and
2. if $a = 1$, then f is affine.

If f is non-affine, the second condition forces $a = 0$. The preserved Paulis are generated by the n input Z operators, X_t , and X_S for $S \in L(f)$. Hence

$$\log |C(U_f)| = n + 1 + \dim L(f),$$

and the formula follows because U_f acts on $n + 1$ qubits. If f is affine, U_f is Clifford and every Pauli is mapped to a Pauli. $\square$

A.2 The affine gap

Proof of Lemma 15. Write $Q = Q_1 \otimes Q_2$ across the system and reference registers. The Choi identity gives

$$\langle J_{U_f} | Q_1 \otimes Q_2 | J_{U_f} \rangle = 2^{-(n+1)} \text{Tr}(U_f^\dagger Q_1 U_f Q_2^T),$$

so every Choi Pauli expectation is, up to the sign that transposition gives a Pauli, a Pauli coefficient of $U_f^\dagger Q_1 U_f$.

Let the X -type input part of Q_1 be S and let $a \in \{0, 1\}$ indicate whether its target part contains Z_t . Conjugation multiplies the target displacement by $\partial_S f$ and the phase by $(-1)^{af}$, as in Lemma 12, so the I_t and X_t sectors of the result each contribute one character sum. Every nonzero Pauli coefficient is therefore, up to phase, of the form

$$\frac{1}{2} \left(\mathbb{E}_x (-1)^{g_1(x) \oplus \ell(x)} \pm \mathbb{E}_x (-1)^{g_2(x) \oplus \ell(x)} \right), \quad (9)$$

where ℓ is affine, $g_1 = af$ and $g_2 = af \oplus \partial_S f$. If $a = 1$, then $g_2(x) = f(x \oplus S)$, so g_1 and g_2 have the same distance from affine functions. If $a = 0$, then $g_1 = 0$ and $g_2 = \partial_S f$.

For any non-affine g and affine ℓ ,

$$|\mathbb{E}(-1)^{g \oplus \ell}| \leq 1 - 2 \text{dist}(g, \text{Aff}).$$

Thus if one of g_1, g_2 is non-affine, (9) has magnitude at most

$$\frac{1}{2}(1 + (1 - 2\delta_f)) = 1 - \delta_f.$$

Two cases remain, and both have g_1, g_2 affine.

- If $\partial_S f$ is constant, then $a = 0$, since $a = 1$ would make $g_1 = f$ non-affine, and $U_f Q_1 U_f^\dagger$ is a single Pauli by [Lemma 12](#). The expectation is then ± 1 , in which case a Hermitian representative of Q stabilizes $|J_{U_f}\rangle$, or 0, which is at most $1 - \delta_f$ because $\delta_f \leq 1/2$.
- If $\partial_S f = g_1 \oplus g_2$ is nonconstant, the two affine characters are distinct, so for a fixed ℓ at most one correlation in (9) is nonzero, and the coefficient is at most $1/2 \leq 1 - \delta_f$.

A coefficient has magnitude one exactly when the conjugated operator is a single Pauli, which is equivalent to a Hermitian representative of Q stabilizing the Choi state. $\square$

A.3 Rigidity of branch nullity and an expected-cost lower bound

Proof of Lemma 16. By Uhlmann's theorem there is an r -qubit pure state $|\chi\rangle$ such that

$$|\langle \Psi | J_{U_f} \otimes \chi \rangle|^2 = F.$$

Write $|\Phi\rangle = |J_{U_f}\rangle \otimes |\chi\rangle$. For any phase-free Pauli in $\text{Stab}(\Psi)$, choose its signed Hermitian representative $P = sQ \otimes R$ fixing $|\Psi\rangle$, where $s \in \{\pm 1\}$ and Q, R are Hermitian Paulis. The orthogonal projector $\Pi_P = (I + P)/2$ fixes $|\Psi\rangle$, so Cauchy–Schwarz gives

$$\begin{aligned} F &= |\langle \Psi | \Pi_P | \Phi \rangle|^2 \leq \langle \Phi | \Pi_P | \Phi \rangle \\ &= \frac{1 + s \langle J_{U_f} | Q | J_{U_f} \rangle \langle \chi | R | \chi \rangle}{2} \leq \frac{1 + |\langle J_{U_f} | Q | J_{U_f} \rangle|}{2}. \end{aligned}$$

Thus $|\langle J_{U_f} | Q | J_{U_f} \rangle| \geq 2F - 1 > 1 - \delta_f$, and [Lemma 15](#) implies that a Hermitian representative of Q stabilizes $|J_{U_f}\rangle$.

Project the phase-free stabilizer group of $|\Psi\rangle$ onto the Choi registers. Its image has size at most $|\text{Stab}(J_{U_f})|$. An element of the kernel is one whose Choi part is trivial, so it has the form $I_{\text{Choi}} \otimes R$ with R a Pauli on the r ancilla qubits. Restricting the kernel to those qubits therefore embeds it in $\mathcal{P}_r$ as a commuting subgroup of signed Paulis that does not contain $-I$, that is, as an isotropic subgroup, and such a subgroup has order at most 2^r . Therefore

$$|\text{Stab}(\Psi)| \leq 2^r |\text{Stab}(J_{U_f})|.$$

Since $|\Psi\rangle$ has $2(n+1) + r$ qubits,

$$\begin{aligned} \nu(\Psi) &\geq 2(n+1) + r - \log(2^r |\text{Stab}(J_{U_f})|) \\ &= \nu(J_{U_f}) = \nu(U_f). \end{aligned}$$

$\square$

Proof of Theorem 17. Run an adaptive implementation on the stabilizer input $|\Omega_{n+1}\rangle$ together with its clean ancillae. Write b for a complete record of the measurements and internal randomness, p_b for its probability, and t_b for the number of T gates used on that branch. Let $|\Psi_b\rangle$ be the pure conditional state on the Choi and retained ancilla registers, before anything is discarded. Since each T gate raises the nullity by at most one and the input is a stabilizer state, $\nu(\Psi_b) \leq t_b$. Put $\rho_b = \text{Tr}_{\text{anc}} |\Psi_b\rangle\langle\Psi_b|$ and $F_b = \langle J_{U_f} | \rho_b | J_{U_f} \rangle$. The averaged reduced state is $J(\mathcal{E})$, so (3) gives

$$\sum_b p_b (1 - F_b) \leq \varepsilon. \quad (10)$$

Call a branch low-nullity if $\nu(\Psi_b) < \nu(U_f)$. By the contrapositive of [Lemma 16](#), every low-nullity branch has

$$1 - F_b \geq \frac{\delta_f}{2}.$$

Therefore (10) implies

$$\Pr_b[\text{low-nullity}] \leq \frac{2\varepsilon}{\delta_f}.$$

Every remaining branch satisfies $t_b \geq \nu(\Psi_b) \geq \nu(U_f)$. Hence the expected number of T gates on the Choi input is at least

$$\mathbb{E}_b[t_b] \geq \left(1 - \frac{2\varepsilon}{\delta_f}\right)_+ \nu(U_f).$$

The branch distribution here is the one obtained from the maximally mixed system input, since the reference is untouched. The worst-input adaptive cost is therefore at least the displayed expectation, by the affineness noted in [Definition 10](#). $\square$

Proof of Lemma 19. Choose $|\Phi\rangle = |J\rangle \otimes |\chi\rangle$ with $|\langle\Psi | \Phi\rangle|^2 = F$ by Uhlmann's theorem. Let G be the group of signed Hermitian Paulis fixing $|\Psi\rangle$. Its elements commute, it does not contain $-I$, and $|G| = 2^{N+r-\nu}$. Hence

$$\Pi_G = \frac{1}{|G|} \sum_{P \in G} P$$

is the orthogonal projector onto their common +1 eigenspace. Since $\Pi_G|\Psi\rangle = |\Psi\rangle$,

$$F \leq \langle\Phi|\Pi_G|\Phi\rangle.$$

Project G onto the phase-free Paulis on the first N qubits. Its kernel consists of signed ancilla-only stabilizers and is an isotropic subgroup on r qubits, so every nonempty fiber has size at most 2^r . If $P = sQ \otimes R \in G$, then

$$\langle\Phi|P|\Phi\rangle \leq |\langle J|Q|J\rangle| |\langle\chi|R|\chi\rangle| \leq |\langle J|Q|J\rangle|.$$

At most $\min\{|G|, 2^r K_\tau\}$ summands can therefore exceed $1 - \tau$, and each is at most 1. Averaging gives

$$F \leq 1 - \tau + \tau \min\left\{1, \frac{2^r K_\tau}{|G|}\right\} = 1 - \tau + \tau \min\{1, K_\tau 2^{v-N}\}.$$

$\square$

A.4 The exact bracket

Proof of Proposition 26. Take the l -th measurement, of qubit q_l . Introduce a fresh ancilla a_l in $|0\rangle$, replace the measurement by a CNOT from q_l to a_l , and retain a_l as a coherent record of the outcome m_l . Every later operation conditioned on that outcome is replaced by the corresponding coherent control on a_l , and a_l is never itself modified. This is the standard principle of deferred measurement.

The deferred circuit is unitary, but for the purpose of analyzing its reduced output channel we may dephase a_l in the computational basis immediately after the copy. Every later use of a_l is diagonal in that basis, and a_l is traced out at the end, so the final partial trace annihilates all cross-outcome coherences and the analytical dephasing does not change the reduced channel. In this dephased description the state is exactly the one produced by the original measurement instrument with its classical outcome stored in a_l . Induction over the circuit therefore gives the joint state of data, working qubits, and record ancillae as $\sum_m p_m \rho_m \otimes |m\rangle\langle m|$, where ρ_m and p_m are the branch states and probabilities of the original circuit.

By [Definition 25](#) the conditioned gates are Paulis applied on affine conditions. A Pauli P applied when $c_0 \oplus \bigoplus_{l \in S} m_l = 1$ becomes $P^{c_0} \prod_{l \in S} C_{a_l}(P)$. That is a product of controlled Paulis, hence Clifford, and on the state above it applies P to exactly the branches meeting the condition.

The result is a unitary Clifford+ T circuit with the same t T gates, ancillae starting in $|0\rangle$ and discarded at the end, whose channel on the data matches the adaptive circuit's. Allowing the classical randomness back in makes it a distribution over such circuits, which is a mixed implementation in the sense of [Definition 9](#), at diamond error ε and cost t . The first claim is then GKZ's bound $\mathcal{T}_\varepsilon^{\text{mixed}}(U_f) \geq \text{RPDT}_\varepsilon^{\text{na}}(f) - 1$ [[GKZ25](#)].

The general form works the same way, except that a conditioned gate may now need T gates. Defer G_i by computing g_i from the record ancillae into a fresh ancilla and applying G_i controlled on it, which adds three costs.

The first cost is computing the condition. An $\{\text{AND}, \oplus, 1\}$ circuit for g_i with $c_\wedge(g_i)$ AND gates becomes a reversible circuit whose CNOT and X gates are Clifford and whose AND gates are exact Toffolis of constant T -count. Running it forwards and then backwards to uncompute costs $O(c_\wedge(g_i))$ T gates, with the record ancillae used only as controls.

The second is applying the gate. A controlled Pauli is Clifford and needs no T gate. A controlled Clifford or controlled T gate on $c + 1$ qubits is a fixed unitary of fixed dimension, and its entries lie in $\mathbb{Z}[1/\sqrt{2}, i]$, so the exact-synthesis theorem of Giles and Selinger [[GS13](#)] makes it exactly representable over Clifford+ T with ancillae. Its T -count is therefore finite. Up to global phase there are finitely many such gates on at most c qubits, so fix one exact circuit for each and let κ_c be the largest of their T -counts together with twice the constant T -count of a Toffoli gate, which covers computing and uncomputing each AND. Then κ_c depends only on c .

The third is reusing the measured qubit. The deferred circuit retains it and can apply the subsequent operations to it. If the original circuit resets that qubit, the reset is a Pauli X conditioned on its own outcome, which is handled at no T cost.

Summing the three costs over the corrections and applying the mixed-model lower bound to the deferred circuit gives the claim. $\square$

A.5 All symmetric Boolean oracles

Proof of [Lemma 28](#). Because f is symmetric, $L(f)$ is invariant under permuting coordinates. The only permutation-invariant subspaces of $\mathbb{F}_2^n$ are

$$\{0\}, \quad \langle \mathbf{1} \rangle, \quad E = \{x : |x| \equiv 0 \pmod{2}\}, \quad \mathbb{F}_2^n.$$

Indeed, if an invariant subspace contains a vector other than 0 or $\mathbf{1}$, subtracting a transposed copy produces some $e_i \oplus e_j$, and permutation invariance then gives all of E ; an additional odd-weight vector gives the full space.

The full space cannot equal $L(f)$, because constant single-coordinate derivatives make f affine. Suppose $E \subseteq L(f)$. Write f_r for the value of f on weight r . For distinct i, j , let r be the weight outside coordinates i, j . Then

$$\partial_{\{i,j\}} f(x) = (1 \oplus x_i \oplus x_j)(f_r \oplus f_{r+2}).$$

This derivative vanishes whenever $x_i \neq x_j$. If it is constant, it must vanish identically, so $f_r = f_{r+2}$ for every r . The weight pattern is then 2-periodic, hence f is an affine function of parity, a contradiction. Thus $L(f)$ is either $\{0\}$ or $\langle \mathbf{1} \rangle$. Apply [Lemma 12](#). $\square$

A.6 The majority gap and constant-error bounds

Proof of [Proposition 32](#). First consider distance to constants. Majority itself is balanced. For every nontrivial derivative, [Lemma 31](#) gives distance at least $c/\sqrt{n}$ from both constants.

Now let $g = \partial_S \text{MAJ}_n$, and let $\ell(x) = \langle T, x \rangle \oplus b$ be a nonconstant affine function, so that T is nonempty. Choose $j \in T$. Pairing each x with $x \oplus e_j$ gives

$$\left| \mathbb{E}_x (-1)^{g(x) \oplus \ell(x)} \right| \leq \Pr_x [\partial_j g(x) = 1]. \quad (11)$$

Discrete derivatives commute, and

$$\partial_j \partial_S f(x) = \partial_j f(x) \oplus \partial_j f(x \oplus S),$$

so

$$\Pr[\partial_j g = 1] \leq 2 \text{Inf}_j(\text{MAJ}_n) = O(n^{-1/2}).$$

Thus g has correlation $O(n^{-1/2})$ with every nonconstant affine function, and its distance from such a function is at least $1/2 - O(n^{-1/2})$. Uniformly for odd $n \geq 5$ the influence is at most $3/8$, so the probability above is at most $3/4$ and this distance is at least $1/8$. The same pairing argument bounds every nonconstant Fourier coefficient of $(-1)^{\text{MAJ}_n}$ by $O(n^{-1/2})$. Therefore the only asymptotically small distances in [Definition 14](#) are distances of derivatives from constants, and these are $\Omega(n^{-1/2})$.

For the matching upper bound, a single-coordinate derivative equals one exactly when the other $n - 1$ bits contain $(n - 1)/2$ ones. Hence

$$\Pr[\partial_j \text{MAJ}_n = 1] = 2^{-(n-1)} \binom{n-1}{(n-1)/2} = \Theta(n^{-1/2}).$$

For $n \geq 5$ this derivative is non-affine, so it appears in the minimum defining δ_{MAJ_n} . $\square$

Proof of Theorem 33. Set $f = \text{MAJ}_n$, $|J\rangle = |J_{U_f}\rangle$, and $N = 2(n+1)$. We bound the number $K_\tau(J)$ of phase-free N -qubit Paulis whose expectation in $|J\rangle$ has magnitude greater than $1 - \tau$, as defined in [Lemma 19](#). Write

$$\phi(x) = (-1)^{f(x)}, \quad D_S(x) = (-1)^{\partial_S f(x)}, \quad p_S = \Pr_x [\partial_S f(x) = 1].$$

Each Choi Pauli expectation is a Pauli coefficient of $U_f^\dagger Q_1 U_f$, up to phase. Parameterize the system Pauli as

$$Q_1 = X^S Z^R \otimes X_t^b Z_t^a, \quad S, R \in \mathbb{F}_2^n, \quad a, b \in \{0, 1\}.$$

If $a = 0$, (9) gives, for an additional input Z -label A , the two target-sector coefficients

$$\frac{\mathbf{1}[A = 0] + \widehat{D}_S(A)}{2}, \quad \frac{\mathbf{1}[A = 0] - \widehat{D}_S(A)}{2},$$

up to phases. At $A = 0$ these are $1 - p_S$ and p_S ; at $A \neq 0$ their magnitudes are at most $1/2$.

If $a = 1$, the two correlations are Fourier coefficients of $\phi(x)$ and $\phi(x \oplus S)$. Since the latter is $(-1)^{A \cdot S} \widehat{\phi}(A)$, each resulting coefficient is zero or has magnitude $|\widehat{\phi}(A)|$. Majority is balanced, so $\widehat{\phi}(0) = 0$. For $A \neq 0$, pairing across any $j \in A$ gives

$$|\widehat{\phi}(A)| \leq \text{Inf}_j(f) = 2^{-(n-1)} \binom{n-1}{(n-1)/2} \leq 3/8,$$

where the last inequality holds for every odd $n \geq 5$.

Because $\tau < 1/4$, a coefficient exceeding $1 - \tau$ in magnitude must therefore be one of $p_S, 1 - p_S$ in the case $a = 0, A = 0$. At most one of these two numbers qualifies. For each fixed S , there are 2^n choices of R and two choices of b , giving

$$K_\tau(J) \leq 2^{n+1} \#\{S : \min(p_S, 1 - p_S) < \tau\}. \quad (12)$$

Odd majority satisfies $\partial_{S^c} f = 1 \oplus \partial_S f$, so $p_{S^c} = 1 - p_S$. Applying [Lemma 31](#) to S and S^c shows that both p_S and $1 - p_S$ are at least $c/2 \geq 2\tau$ whenever $n/4 \leq |S| \leq 3n/4$. The standard binomial entropy bound gives

$$\#\{S : |S| < n/4 \text{ or } |S| > 3n/4\} \leq 2^{h_2(1/4)n+1}.$$

Thus (12) implies

$$K_\tau(J) \leq 2^{n+1+h_2(1/4)n+1} = 2^{N-\alpha n}. \quad (13)$$

Let $|\Psi\rangle$ be any pure state on the Choi registers and any number of additional qubits, and let F_Ψ denote its reduced fidelity with $|J\rangle$. By Lemma 19 and (13),

$$\nu(\Psi) \leq \alpha n - 1 \implies F_\Psi \leq 1 - \tau/2.$$

Run an adaptive ε -implementation on the Choi input. This instantiates Corollary 20 at $\ell_* = \alpha n - 1$. A pure branch b with T -count t_b satisfies $\nu(\Psi_b) \leq t_b$, and (10) gives $\sum_b p_b(1 - F_b) \leq \varepsilon$. Put $B = (\alpha n - 1)_+$. If $B > 0$, every branch with $t_b < B$ has infidelity at least $\tau/2$, so its total probability is at most $2\varepsilon/\tau$. The other branches cost at least B , proving

$$\mathbb{E}_b[t_b] \geq (1 - 2\varepsilon/\tau)_+ B.$$

For $B = 0$ this inequality is trivial. As in Theorem 17, the Choi-input expected cost is at most the worst-input expected cost, proving (4).

For $n \geq 11$, we have $\alpha n - 1 \geq \alpha n/2$, so $\varepsilon \leq \tau/4$ gives the lower bound $\alpha n/4$. To cover $n = 5, 7, 9$, note that an exhaustive computation over f , all its derivatives and all affine functions in Definition 14 gives $\delta_{\text{MAJ}_5} = 1/8$, $\delta_{\text{MAJ}_7} = 3/16$ and $\delta_{\text{MAJ}_9} = 29/128$. Each equals $\frac{1}{2} - 2^{1-n} \binom{n-1}{(n-1)/2}$ and is attained by a single-coordinate derivative and an affine function of the other $n - 1$ bits. Set

$$\varepsilon_* = \tau/4 \leq 1/32.$$

Then $\varepsilon_* \leq \delta_{\text{MAJ}_n}/4$ at these arities, and Theorem 17 gives $\mathcal{T}_\varepsilon^{\text{adaptive}} \geq n/2$. Finally, exact synthesis from Lemma 27 gives $\mathcal{T}_\varepsilon^{\text{unitary}}(U_{\text{MAJ}_n}) = O(n)$, and (1) places the other two costs between the two bounds. $\square$

B Binomial estimates for majority derivatives

We prove Lemma 31. Let $R_r = \xi_1 + \dots + \xi_r$ for independent uniform signs $\xi_i \in \{\pm 1\}$. We use the following elementary local estimate.

Lemma 67 (Local binomial estimate). *There are constants $c, C > 0$ such that for every $r \geq 1$ and every integer t in the support of R_r with $|t| \leq 3\sqrt{r}$,*

$$\frac{c}{\sqrt{r}} \leq \Pr[R_r = t] \leq \frac{C}{\sqrt{r}}.$$

Consequently there is an absolute r_0 such that for every $r \geq r_0$,

$$\begin{aligned} \Pr\left[\frac{1}{2}\sqrt{r} \leq |R_r| \leq \sqrt{r}\right] &\geq c, \\ \Pr\left[2\sqrt{r} \leq |R_r| \leq 3\sqrt{r}\right] &\geq c. \end{aligned}$$

The threshold is needed: below it the second interval may contain no admissible lattice point at all, as at $r = 3$, where $|R_3| \in \{1, 3\}$ misses $[2\sqrt{3}, 3\sqrt{3}]$. The proof of Lemma 31 handles the finitely many small cases separately.

Proof. The central mass satisfies

$$2^{-r} \binom{r}{\lfloor r/2 \rfloor} = \Theta(r^{-1/2}).$$

For a supported value $t = 2d$ or $2d + 1$ according to parity, the ratio between the mass at t and the central mass is a product of $O(|d|)$ factors of the form

$$\frac{r/2 - i + O(1)}{r/2 + i + O(1)}.$$

For all sufficiently large r , when $|t| \leq 3\sqrt{r}$, the logarithm of this product is bounded above and below by absolute constants, because $\sum_{i \leq O(\sqrt{r})} i/r = O(1)$. This proves the pointwise estimate for all sufficiently large r . Decreasing c and increasing C to cover the finitely many smaller values of r proves it for every $r \geq 1$. Each displayed interval contains $\Theta(\sqrt{r})$ admissible lattice points for all sufficiently large r , so summing their probabilities gives the stated constant lower bounds. $\square$

Proof of Lemma 31. Let $s = |S|$ and $m = n - s$. Encode the input bits by independent signs, and let $A = R_s$ and $B = R_m$ be the signed sums on S and its complement. Flipping the coordinates in S changes A to $-A$ and leaves B fixed. Since n is odd, neither $A + B$ nor $B - A$ is zero, and majority changes value exactly when

$$(A + B)(B - A) < 0, \quad \text{equivalently} \quad |B| < |A|. \quad (14)$$

Denote this probability by p_s .

Assume first $1 \leq s \leq n/2$, so $m \geq s$. For all sufficiently large s , Lemma 67 gives constant probability that

$$\frac{1}{2}\sqrt{s} \leq |A| \leq \sqrt{s}.$$

Conditioned on any such A , the interval $(-|A|, |A|)$ contains $\Theta(|A|)$ admissible values of B , each within $\sqrt{m}$ of the origin and hence each having probability $\Omega(m^{-1/2})$. Therefore $p_s \geq c\sqrt{s/m} \geq c\sqrt{s/n}$. For bounded s , choose any fixed nonzero admissible value of A occurring with constant probability and one admissible central value of B with smaller magnitude, which exists because A and B have opposite parities when n is odd. The local estimate gives $p_s \geq c/\sqrt{m} \geq c/\sqrt{n}$. Thus

$$p_s \geq c \frac{\max\{1, \sqrt{s}\}}{\sqrt{n}} \quad (1 \leq s \leq n/2). \quad (15)$$

We also need p_s bounded away from one in this range. With constant probability $|A| \leq \sqrt{s}$, by Lemma 67 for $s \geq r_0$ and because $A \in \{0, \pm 1\}$ has positive probability for smaller s , while independently, with constant probability,

$$2\sqrt{m} \leq |B| \leq 3\sqrt{m}.$$

On their intersection $|B| > |A|$, so (14) fails. Hence $p_s \leq 1 - c$ once $m \geq r_0$.

The remaining pairs (n, S) are those with $m = n - s < r_0$, and since $s \leq n/2$ forces $m \geq n/2$, they all have $n < 2r_0$: finitely many. For each of them $\partial_S \text{MAJ}_n$ is nonconstant, because a nonempty proper S leaves some input at which flipping the coordinates of S crosses the majority threshold and some input at which it does not, so $p_s \in (0, 1)$ strictly. Taking c no larger than the minimum of $\min\{p_s, 1 - p_s\}$ over these finitely many pairs covers them, both for the bound $c/\sqrt{n}$ and for the constant bound used below.

For $s > n/2$, odd-arity majority obeys $\text{MAJ}_n(x \oplus \mathbf{1}) = 1 \oplus \text{MAJ}_n(x)$, and therefore

$$\partial_{S^c} \text{MAJ}_n = 1 \oplus \partial_S \text{MAJ}_n, \quad p_s = 1 - p_{n-s}.$$

Combining this identity with (15) and the constant upper bound proves $\min\{p_s, 1 - p_s\} \geq c/\sqrt{n}$ for every nonempty proper S .

Finally, if $\zeta n \leq s \leq n/2$, then (15) gives $p_s \geq c\sqrt{\zeta}$. If $n/2 < s \leq (1 - \zeta)n$, then $n - s \leq n/2$ and $p_s = 1 - p_{n-s} \geq c \geq c\sqrt{\zeta}$. This proves the second claim. $\square$

C Proofs of the workspace results

C.1 Fourier sampling produces a signed threshold

Proof of Theorem 34. Fix x and put $Y_j = L\sigma_j \chi_{S_j}(x)$. These are independent and lie in $[-L, L]$, and each one has expectation $f(x)$:

$$\mathbb{E}[Y_j] = L \sum_S \frac{|\widehat{f}(S)|}{L} \text{sign}(\widehat{f}(S)) \chi_S(x) = \sum_S \widehat{f}(S) \chi_S(x) = f(x).$$

If rounding their average at $1/2$ gives the wrong bit, that average differs from $f(x)$ by at least $1/2$. Hoeffding's inequality therefore gives

$$\Pr[g(x) \neq f(x)] \leq 2 \exp\left(-\frac{k}{8L^2}\right) \leq \frac{\varepsilon}{4}.$$

This holds for every x . For each sampled branch,

$$U_g = U_f(I - 2\Pi_g), \quad \Pi_g = \sum_{x: g(x) \neq f(x)} |x\rangle\langle x| \otimes |-\rangle\langle -|.$$

Thus $\|\mathbb{E}\Pi_g\|_\infty = \max_x \Pr[g(x) \neq f(x)]$. Applying the general reflection bound in [Lemma 11](#) proves the channel estimate. $\square$

C.2 A blocked reversible tally

Proof of Proposition 35. Fix one sample $S_1, \dots, S_k$ and evaluate the comparison in (7) exactly. Let

$$\beta = \max\{2, \lceil \log k \rceil\}$$

and divide the k positions into consecutive blocks of at most β positions. Maintain an accumulator A of $\lceil \log(k+1) \rceil$ qubits, initially zero.

For one block, compute its parity values z_j into at most β clean qubits using CNOT gates, and apply X to the coordinates with $c_j = 1$. Reversibly count how many of these β bits are set, writing the total into a block-sum register of $\lceil \log(\beta+1) \rceil$ qubits, enough to hold any count from 0 to β . A balanced tree of reversible adders uses $O(\beta)$ Toffoli gates and $O(\beta)$ temporary qubits: at level r there are $O(\beta/2^r)$ additions of $O(r)$ -bit numbers, and

$$\sum_{r \geq 0} O\left(\frac{\beta r}{2^r}\right) = O(\beta).$$

Add the block sum into A with a Toffoli-based ripple-carry adder using $O(\log k)$ Toffoli gates. Then reverse the counting circuit and the parity computations, clearing all block workspace before processing the next block.

There are $O(k/\beta) = O(k/\log k)$ blocks. Each block costs $O(\beta + \log k) = O(\log k)$ Toffoli gates, so the forward accumulation costs $O(k)$. After the last block, compare A with the constant θ using $O(\log k)$ Toffoli gates, copy the comparison bit to the target, uncompute the comparison, and reverse the accumulation. The total number of Toffoli gates, hence the total T -count, is $O(k)$.

At any time, the live clean registers are the $O(\log k)$ -qubit accumulator, one $O(\log k)$ -bit block, and $O(\log k)$ temporary adder and comparator qubits. Thus $\lambda = O(\log k)$. The branch computes U_g exactly, so averaging the branches gives the error guarantee of [Theorem 34](#). $\square$

C.3 One clean ancilla with a read-only input

Proof of Proposition 36. Apply [Theorem 34](#) with sampling error $\varepsilon/2$, so that

$$k = \left\lceil 8L^2 \ln \frac{16}{\varepsilon} \right\rceil = \Theta\left(L^2 \log(8/\varepsilon)\right), \quad L = \|\hat{f}\|_1,$$

and allocate the remaining error budget to compiling the resulting branch. This k is larger than the one in [Proposition 35](#) by the halved error budget, and of the same order. For a fixed sample, the top function is a threshold of k parities of n input bits. Each parity is an XOR tree of depth $O(\log n)$, and a threshold of k bits has a formula of depth $O(\log k)$ built from carry-save adders [PPZ92], so the top function has formula depth $d = O(\log(nk))$. Barrington's theorem [Bar89] gives a width-5 permutation branching program over A_5 of length $4^{O(d)} = \text{poly}(n, k)$ whose product is the identity when $g(x) = 0$ and a fixed nonidentity element when

$g(x) = 1$. Since A_5 is simple, some fixed word in conjugates of that element and its inverse equals the double transposition $v = (12)(34)$. Concatenating conjugated copies of the program, reversed where an inverse is needed, realizes that word at constant-factor cost in length. Instructions are applied in the reverse of program order, so that the composed unitaries multiply in the order the program does.

Let $\rho : A_5 \rightarrow O(4)$ be the standard representation: the permutation action on five points with the trivial subrepresentation removed. The element v has eigenvalues $(1, 1, -1, -1)$ under ρ . After a fixed orthogonal change of basis, we may therefore assume $\rho(v) = X \otimes I$ on a two-qubit register, identified with the target and one clean ancilla.

An instruction of the branching program has the form (j, π_0, π_1) : query x_j and multiply by π_{x_j} . Coherently implement it by applying the controlled two-qubit unitary

$$|0\rangle\langle 0|_{x_j} \otimes I + |1\rangle\langle 1|_{x_j} \otimes \rho(\pi_0^{-1}\pi_1)$$

and then the fixed $\rho(\pi_0)$. Since ρ is a homomorphism, the composite is $\rho(\pi_0)\rho(\pi_0^{-1}\pi_1) = \rho(\pi_1)$ when $x_j = 1$ and $\rho(\pi_0)$ when $x_j = 0$, as required. The input qubit is used only as a control. The product over all instructions acts on the target–ancilla pair as I when $g(x) = 0$ and as $X \otimes I$ when $g(x) = 1$. Hence it implements U_g coherently and returns the ancilla to its initial state. The compilation step below preserves this up to the approximation error.

There are $L_B = \text{poly}(n, k)$ fixed controlled unitaries of constant dimension, compiled as follows. Every element of A_5 is an even permutation, so $\det \rho(\pi) = \text{sign}(\pi) = 1$ and each $\rho(\pi)$ lies in $SO(4)$. Let $\mathcal{M}$ be the magic basis, which can be chosen as a Clifford change of coordinates acting on the two workspace qubits alone, under which $SO(4)$ becomes $SU(2) \otimes SU(2)$ [VW04], and write $\rho(\pi) = \mathcal{M}^\dagger (P_\pi \otimes Q_\pi) \mathcal{M}$ with $P_\pi, Q_\pi \in SU(2)$. The controlled instruction then factors as

$$|0\rangle\langle 0|_{x_j} \otimes I + |1\rangle\langle 1|_{x_j} \otimes \rho(\pi) = (I \otimes \mathcal{M}^\dagger) \Lambda(P_\pi) \Lambda(Q_\pi) (I \otimes \mathcal{M}),$$

where $\Lambda(\cdot)$ denotes control on x_j and $\mathcal{M}, \mathcal{M}^\dagger$ do not touch the input. Each $\Lambda(P_\pi)$ is a controlled single-qubit gate, so the standard ABC decomposition [BBC⁺95] applies, and since $\det P_\pi = 1$ the phase gate it would otherwise place on the control is trivial. Approximating the one-qubit factors A, B, C by Solovay–Kitaev compilation [DN06] to operator-norm error $O(\varepsilon/L_B)$ costs $\text{polylog}(L_B/\varepsilon)$ Clifford+ T gates each, while the CNOTs stay exact, so the input qubit remains a control throughout. Choosing the compilation constant sufficiently small makes the accumulated branch error at most $\varepsilon/2$ and the total T -count is $L_B \text{polylog}(L_B/\varepsilon) = \text{poly}(n, k)$, because $k = \Omega(\log(1/\varepsilon))$. The ancilla returns to $|0\rangle$ up to this compilation error. Averaging over the sampled branches completes the construction. $\square$

Proof of Proposition 39. Since every input wire is used only as a control, the circuit has the form

$$V = \sum_x |x\rangle\langle x| \otimes V_x.$$

Let m_j be the number of CNOTs controlled by x_j , reduced modulo 2. Multiplicativity of the determinant gives

$$\det V_x = c (-1)^{\sum_j m_j x_j},$$

where c is the product of the determinants of all target-only gates and is independent of x . Exact coherent synthesis requires $V_x = e^{i\phi} X^{g(x)}$ for one fixed ϕ , so dividing the determinant identity at x by its value at 0 gives

$$(-1)^{g(x) \oplus g(0)} = (-1)^{\sum_j m_j x_j}.$$

Hence $g(x) = g(0) \oplus \bigoplus_j m_j x_j$. Conversely, an affine oracle uses only target X gates and the indicated CNOTs. $\square$

C.4 Randomized Toffoli from a linear-subspace test

Proof of Proposition 41. Take the branch of Construction 40, so $\ell \leq n - 1$. Fix a nonzero $y \in \mathbb{F}_2^{n-1}$. Under the uniform distribution on independent ordered ℓ -tuples,

$$\Pr[a_j \cdot y = 0 \ \forall j] = \frac{\prod_{i=0}^{\ell-1} (2^{n-2} - 2^i)}{\prod_{i=0}^{\ell-1} (2^{n-1} - 2^i)} \leq 2^{-\ell}.$$

That probability is what the error operator measures: the projector Π_a is diagonal in the computational basis of the data, tensored with the target state $|-\rangle$, its diagonal entry at a nonzero y is exactly the event just bounded, and it vanishes at $y = 0$. Averaging over a therefore gives $\|\mathbb{E}_a \Pi_a\|_\infty \leq 2^{-\ell}$, and Lemma 11 converts this into a diamond error of at most $4 \cdot 2^{-\ell} \leq \varepsilon$.

It remains to count the gates. The CNOT basis change and the negative controls are Clifford, so every T gate is in the ℓ -controlled NOT. Barenco et al. decompose that gate exactly using one extra qubit and $O(\ell)$ three-qubit Toffolis [BBC⁺95, Cor. 7.4], and a three-qubit Toffoli has constant T -count, so the branch uses $O(\ell)$ T gates. (The corollary is stated for networks of at least seven bits, which here means $\ell \geq 5$; for $\ell \leq 4$ a fixed exact decomposition of $\Lambda_\ell(X)$ that uses the same work qubit, which may be borrowed, serves at constant cost.)

One extra qubit suffices because the decomposition splits the controls into two halves and lets each half act as the work register for the other. This is legitimate because the work bits in [BBC⁺95, Lem. 7.2 and the remark following it] are correct “independent of the initial state of the bits” and are returned to their initial values, so the controls are borrowed as dirty scratch and restored. The same reasoning means the extra qubit may be dirty. When $\ell \leq n - 2$, at least one transformed data coordinate lies outside the ℓ controls; borrow that coordinate as the extra work qubit and restore it before undoing the basis change. This uses no clean ancilla. When every transformed data coordinate is a control, supply one clean extra qubit. In either case, each branch is an exact Boolean oracle and its work qubits are restored exactly.

If $\ell > n - 1$, use the same Barenco construction to implement Toff_n exactly with one clean ancilla and $O(n)$ T gates; the conditionally clean ancilla technique of Khattar and Gidney [KG24] reaches the same regime with better constants. Combining the regimes gives the stated upper bound. GKZ prove the stronger adaptive statement $\mathcal{T}_\varepsilon^{\text{mixed}}(\text{Toff}_n) \geq \mathcal{T}_\varepsilon^{\text{adaptive}}(\text{Toff}_n) = \Omega(\min\{n, \log(1/\varepsilon)\})$, for large enough n and $1/\varepsilon$ [GKZ25], so the upper bound above is optimal up to constants in that regime. $\square$

C.5 Bounded spectral norm and a reusable clean ancilla

Proof of Proposition 43. For $f \equiv 0$ use the identity circuit. Otherwise take a decomposition (8) with $1 \leq J \leq q(M)$, and write $h_j = \mathbf{1}_{a_j + V_j}$. Reducing the integer identity modulo 2 removes the signs and gives

$$f(x) = \bigoplus_{j=1}^J h_j(x), \quad U_f = U_{h_J} \cdots U_{h_1}.$$

A coset of codimension r has a description $a + V = \{x : Bx = b\}$ with B of full row rank r . Complete its rows to a basis and implement the resulting invertible linear transformation by CNOTs. An X layer then changes the first r coordinates so that $Bx = b$ is equivalent to all of them being 1.

If $r = 0$, the coset indicator is constant and its oracle is a target X ; if $r = 1$, its oracle is affine and uses only Clifford gates. If $r \geq 2$, apply Proposition 41 to the gate Toff_{r+1} on these r coordinates and the target, with error budget ε/J , and undo the affine coordinate change. The resulting coset-oracle channel $\mathcal{E}_j$ has error at most ε/J and uses $O(\log(2J/\varepsilon))$ T gates per branch and at most one clean ancilla.

Each constituent branch returns the clean ancilla exactly to $|0\rangle$, so the same qubit can be reused for all J factors. Choose the factors’ classical randomness independently. The resulting averaged channel is $\mathcal{E}_J \circ \cdots \circ \mathcal{E}_1$,

and the triangle inequality and contractivity of diamond distance under channel composition give

$$D_{\diamond}(\mathcal{E}_J \circ \cdots \circ \mathcal{E}_1, U_f) \leq \sum_{j=1}^J D_{\diamond}(\mathcal{E}_j, U_{h_j}) \leq \varepsilon.$$

The worst-branch T -count is $O(J \log(2J/\varepsilon))$, bounded by the stated expression. The clean ancilla is restored exactly in every joint branch. $\square$

C.6 Exact single-pass evaluation

Proof of Proposition 45. First suppose $\theta \leq (k+1)/2$. For each $s \in \{0, 1, \dots, \theta\}$, choose a prefix on the first θ read positions containing exactly s ones, and let ρ_s be the reduced workspace state after that prefix. Fix $s < s'$. Append the same suffix containing exactly $\theta - s'$ ones. This is possible because $s' \geq 1$, and hence $0 \leq \theta - s' \leq \theta - 1 \leq k - \theta$. The two completed inputs have weights $s + \theta - s' < \theta$ and $s' + \theta - s' = \theta$, so their outputs must be 0 and 1.

The remainder of the evaluator induces the same quantum channel on the workspace for both prefixes: the already-read positions are never touched again, and the suffix is identical. The two final states are perfectly distinguished by the output measurement. Trace distance is contractive, so $D(\rho_s, \rho_{s'}) = 1$, equivalently their supports are orthogonal. Thus $\rho_0, \dots, \rho_{\theta}$ have $\theta + 1$ pairwise orthogonal supports, forcing $\dim W \geq \theta + 1$.

If $\theta > (k+1)/2$, complement every input bit and negate the output. This turns THR_{θ} into $\text{THR}_{k-\theta+1}$ without changing the workspace, and the first case gives dimension at least $k - \theta + 2$.

The bound is attained in this channel model. For $\theta \leq (k+1)/2$, retain the classical saturated count $c \in \{0, \dots, \theta\}$ and update $c \leftarrow \min\{c + v_i, \theta\}$. Measuring this classical register and updating it defines a channel, which Definition 44 permits; the final answer is $\mathbf{1}[c = \theta]$. For larger θ , apply the same construction after complementing the input and output. The required dimension is exactly $\min\{\theta + 1, k - \theta + 2\}$. $\square$

C.7 Single-pass evaluation at bounded error

Proof of Lemma 47. The case $M = 1$ is immediate. The decoding is a binary search: each threshold query resolves one bit of the index. Put $r = \lfloor \log M \rfloor$, choose a uniformly random string $Z = Z_1 \cdots Z_r \in \{0, 1\}^r$, and encode it as $\rho_{\text{int}(Z)}$, with the most significant bit first. For each i and each prefix of integer value $p = \text{int}(Z_1 \cdots Z_{i-1})$, set

$$t(p, i) = p 2^{r-i+1} + 2^{r-i}.$$

This is the midpoint of the interval the first $i-1$ bits leave open, and lies in $\{1, \dots, 2^r - 1\} \subseteq \{1, \dots, M-1\}$. Conditioned on that prefix, $\text{int}(Z) \geq t(p, i)$ holds exactly when $Z_i = 1$. Thus measuring $E_{t(p, i)}$ recovers Z_i with error at most η . Conditional Fano's inequality and data processing give

$$I(Z_i : W \mid Z_1 \cdots Z_{i-1}) \geq 1 - h_2(\eta).$$

Applying the chain rule to the classical-quantum encoding, followed by the Holevo bound, yields

$$r(1 - h_2(\eta)) \leq \sum_{i=1}^r I(Z_i : W \mid Z_1 \cdots Z_{i-1}) = I(Z : W) \leq \log D.$$

This is the serial-encoding form of the random-access-code bound. $\square$

Proof of Theorem 48. As in Proposition 45, complementing inputs and output lets us assume that $\theta = m \leq (k+1)/2$. For every $s \in \{0, \dots, m\}$, choose a prefix on the first m positions of weight s , and let ρ_s be its reduced workspace state. For each $t \in \{1, \dots, m\}$, choose a common suffix of weight $m - t$. These suffixes exist because $0 \leq m - t \leq m - 1 \leq k - m$.

The continuation and final measurement induce an effect E_t on the workspace. No previously read position or discarded environment is accessible to that continuation. The completed input has weight $s + m - t$, so correctness gives

$$|\text{Tr}(E_t \rho_s) - \mathbf{1}[s \geq t]| \leq \eta.$$

Apply [Lemma 47](#) with $M = m + 1$, and use $\log \dim W \leq w$. $\square$

C.8 Transfer to sampled branches

Proof of [Lemma 51](#). After $j < k$ samples, their span has dimension at most j . The next sample falls in that span with probability at most γ/k . A union bound gives failure probability at most γ .

Conditioning a probability distribution on an event of probability $1 - q$ changes it by total variation distance exactly q . If Φ_a denotes the channel associated with sample tuple a , then

$$D_\diamond(\mathcal{E}, \mathcal{E}') \leq \frac{1}{2} \sum_a |p(a) - p'(a)| \|\Phi_a\|_\diamond = \text{TV}(p, p') = q \leq \gamma,$$

since every channel has diamond norm one. $\square$

Proof of [Corollary 52](#). When $S_1, \dots, S_k$ are linearly independent, the linear map

$$x \mapsto (\langle S_1, x \rangle, \dots, \langle S_k, x \rangle)$$

is surjective onto $\mathbb{F}_2^k$. The fixed sign corrections $b_j = \langle S_j, x \rangle \oplus c_j$ preserve surjectivity. Thus every prefix and suffix used in [Proposition 45](#) and [Theorem 48](#) is realized by an actual input x .

For the error bound, let E be the event of linear independence. By [Lemma 51](#), $\Pr(E) \geq 1 - \gamma > 0$. The pointwise guarantee (6) gives

$$\Pr[g(x) \neq f(x) \mid E] \leq \frac{\Pr[g(x) \neq f(x)]}{\Pr(E)} \leq \frac{\varepsilon}{4(1 - \gamma)} \quad \text{for every } x.$$

For Boolean XOR oracles, $U_g = U_f(I - 2\Pi_g)$ with

$$\Pi_g = \sum_{x: g(x) \neq f(x)} |x\rangle\langle x| \otimes |-\rangle\langle -|.$$

Therefore $\|\mathbb{E}[\Pi_g \mid E]\|_\infty \leq \varepsilon/[4(1 - \gamma)]$, and [Lemma 11](#) yields the stated diamond error. The additive γ estimate in [Lemma 51](#) is a general distributional bound; the pointwise oracle structure gives this stronger multiplicative bound. $\square$

C.9 Clean versus borrowed workspace

Proof of [Proposition 56](#). Write $|\psi_r\rangle = V^r|\psi\rangle$. Every $r \in \{0, \dots, N\}$ is realized by some input, so the fixed effect E must label $|\psi_r\rangle$ by $\mathbf{1}[r \geq \theta]$.

For each $d \in \{1, \dots, N\}$, set $r = \max\{0, \theta - d\}$ and $s = r + d$. Then $0 \leq r < \theta \leq s \leq N$. In the exact case, $|\psi_r\rangle$ and $|\psi_s\rangle$ are perfectly distinguished by E and are therefore orthogonal. Unitarity gives

$$0 = \langle \psi_r \mid \psi_s \rangle = \langle \psi \mid V^d \mid \psi \rangle.$$

It follows that all $N + 1$ vectors $|\psi_0\rangle, \dots, |\psi_N\rangle$ are pairwise orthogonal, proving $D \geq N + 1$. The stated cyclic shift attains the bound.

For error at most $\eta < 1/2$, the acceptance probabilities of the same pair differ by at least $1 - 2\eta$. Their trace distance is consequently at least $1 - 2\eta$, and the formula for the trace distance between pure states yields

$$|\langle \psi | V^d | \psi \rangle|^2 = |\langle \psi_r | \psi_s \rangle|^2 \leq 1 - (1 - 2\eta)^2 = 4\eta(1 - \eta).$$

Thus the Gram matrix G of $|\psi_0\rangle, \dots, |\psi_N\rangle$ has unit diagonal and squared moduli of off-diagonal entries at most $4\eta(1 - \eta)$. Since G is positive semidefinite,

$$D \geq \text{rank } G \geq \frac{(\text{Tr } G)^2}{\text{Tr}(G^2)} \geq \frac{N + 1}{1 + 4N\eta(1 - \eta)}.$$

For the entropy bound, replace $(V, |\psi\rangle, E)$ by $(V^{-1}, V^N |\psi\rangle, I - E)$ if necessary. The resulting acceptance probabilities approximate the threshold at $N + 1 - \theta$, so we may assume $\theta = m \leq (N + 1)/2$. For $a = 0, \dots, m$, let $\rho_a = V^a |\psi\rangle\langle\psi| V^{-a}$; for $t = 1, \dots, m$, let

$$E_t = V^{-(m-t)} E V^{m-t}.$$

Then $\text{Tr}(E_t \rho_a)$ is the acceptance probability at count $a + m - t$, which lies in $\{0, \dots, 2m - 1\} \subseteq \{0, \dots, N\}$. It therefore approximates $\mathbf{1}[a \geq t]$ with error at most η . Applying the serial-encoding bound of [Lemma 47](#) to these $m + 1$ states proves the second inequality. $\square$

Proof of Proposition 57. For $S \subseteq [k]$, write $C_S = \prod_{j \in S} C_j$. Given any nonempty S , choose $R \subseteq [k] \setminus S$ with $|R| = \max\{0, \theta - |S|\}$. Such an R exists because $\theta \leq k$. The indicator strings of R and $R \cup S$ have opposite threshold values. Since the gates commute, their states are $C_R |\psi\rangle$ and $C_R C_S |\psi\rangle$. Exactness therefore gives $\langle \psi | C_S |\psi\rangle = 0$.

Consider the $k + 1$ chain states

$$|\chi_a\rangle = C_{\{1, \dots, a\}} |\psi\rangle, \quad 0 \leq a \leq k.$$

For $a < b$, their inner product is $\langle \psi | C_{\{a+1, \dots, b\}} |\psi\rangle$, which vanishes. Thus they are pairwise orthogonal and $D \geq k + 1$. Taking every C_j to be the cyclic shift on $k + 1$ states attains the bound.

With error at most η , the opposite-label pair constructed above instead gives $|\langle \psi | C_S |\psi\rangle|^2 \leq 4\eta(1 - \eta)$ by the same pure-state trace-distance argument as in [Proposition 56](#). All distinct chain states have squared overlap bounded by this quantity. The Gram-matrix rank argument consequently gives $D \geq (k + 1)/(1 + 4k\eta(1 - \eta))$.

For the entropy bound, complement all input bits and the output if necessary: replacing C_j by C_j^{-1} , $|\psi\rangle$ by $C_{[k]} |\psi\rangle$, and E by $I - E$ changes the threshold to $k + 1 - \theta$. We may therefore assume $\theta = m \leq (k + 1)/2$. Partition the positions into a set A of size m and a set B of size $k - m \geq m - 1$. Choose $S_a \subseteq A$ of size a for $a = 0, \dots, m$, and $T_t \subseteq B$ of size $m - t$ for $t = 1, \dots, m$. Define

$$\rho_a = C_{S_a} |\psi\rangle\langle\psi| C_{S_a}^\dagger, \quad E_t = C_{T_t}^\dagger E C_{T_t}.$$

Commutativity identifies $\text{Tr}(E_t \rho_a)$ with the acceptance probability on the indicator string of $S_a \cup T_t$. Its weight is $a + m - t$, so this probability approximates $\mathbf{1}[a \geq t]$ with error at most η . Apply [Lemma 47](#) with $m + 1$ states. $\square$

C.10 A pass-workspace hierarchy

Proof of Theorem 62. Give Alice the positions in A and Bob those in B , and let each simulate the evaluator on their own side. Neither can read the other's bits, so whenever the read schedule crosses the cut the current workspace register must be sent across; over c crossings that is at most cw qubits of communication.

The fixed initial workspace state is known, so whichever party performs the first read prepares the whole register locally. The parties start without shared entanglement or shared randomness, as [Lemma 60](#) requires. The protocol computes M_θ exactly. By [Lemmas 60](#) and [61](#),

$$cw \geq \frac{1}{2} \log \text{rank}(M_\theta) \geq \frac{1}{2} \log(k/4 - 1).$$

For a p -pass scan, $c \leq 2p - 1$, which yields $wp = \Omega(\log k)$. $\square$

Proof of [Lemma 63](#). The inverse of Λ_N is bidiagonal, with 1 on the diagonal and -1 on the superdiagonal, so $\Lambda_N^{-1} = I - S$ for the shift S . The shift has operator norm at most 1, so $\|\Lambda_N^{-1}\|_{\text{op}} \leq 2$ and therefore $\sigma_{\min}(\Lambda_N) = \|\Lambda_N^{-1}\|_{\text{op}}^{-1} \geq 1/2$. By Weyl's inequality for singular values, and since the operator norm is dominated by the Frobenius norm,

$$\sigma_{\min}(A) \geq \sigma_{\min}(\Lambda_N) - \|A - \Lambda_N\|_{\text{op}} \geq \frac{1}{2} - \|A - \Lambda_N\|_F > 0,$$

so A is nonsingular. An entrywise bound of $1/(2N)$ gives $\|A - \Lambda_N\|_F < 1/2$. $\square$

Proof of [Corollary 65](#). The argument is the one in [Theorem 62](#), with the exact rank replaced by a perturbed one. Simulating the evaluator gives a two-party protocol without prior entanglement or shared randomness that communicates $q \leq cw$ qubits and accepts (x, y) with probability within η of $\mathbf{1}[|x| + |y| \geq \theta]$. Let A be its acceptance-probability matrix; by [Lemma 60](#), $\text{rank}(A) \leq 4^q$.

Put $k_1 = \lfloor k/2 \rfloor$, $k_2 = \lceil k/2 \rceil$, and $a_0 = \max\{0, \theta - k_2\}$. For every $a \in \{a_0, \dots, a_0 + m - 1\}$ choose one string $x_a \in \{0, 1\}^{k_1}$ of weight a , and for every a'' in the same range choose one string $y_{a''} \in \{0, 1\}^{k_2}$ of weight $\theta - a''$. These strings exist by the weight-range check in the proof of [Lemma 61](#). Let $B[a, a''] = A[x_a, y_{a''}]$. Then B is a submatrix of A , so $\text{rank}(B) \leq \text{rank}(A)$, and its ideal entries are $\mathbf{1}[a + (\theta - a'') \geq \theta] = \mathbf{1}[a \geq a'']$. Thus B^T differs from Λ_m entrywise by at most $\eta < 1/(2m)$, so [Lemma 63](#) gives $\text{rank}(B) = \text{rank}(B^T) = m$. Hence $4^{cw} \geq m$. $\square$

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We used LLMs to assist with expository drafting, technical auditing, and mathematical revisions of this manuscript. Their contributions included identifying model and scope issues, developing the projector-based gap and Pauli-counting arguments, applying serial-encoding bounds, and helping incorporate the coset-decomposition synthesis consequence of the Green–Sanders theorem. These changes affected the introduction, technical overview, and the proofs and statements in the appendices.

The tools also assisted with cross-reference checks, compilation, and comparisons with cited sources. The authors take full responsibility for all mathematical claims, interpretations of prior work, and presentation choices.

References

- [ADOY24] Anurag Anshu, Yangjing Dong, Fengning Ou, and Penghui Yao. On the computational power of QAC^0 with barely superlinear ancillae. arXiv:2410.06499, 2024.
- [AGK⁺05] Farid Ablayev, Aida Gainutdinova, Marek Karpinski, Cristopher Moore, and Christopher Pollett. On the computational power of probabilistic and quantum branching programs. *Information and Computation*, 203(2):145–162, 2005.
- [AM86] Noga Alon and Wolfgang Maass. Meanders, Ramsey theory and lower bounds for branching programs. In *27th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 410–417, 1986.

- [AM19] Matthew Amy and Michele Mosca. T-Count Optimization and Reed–Muller Codes. *IEEE Transactions on Information Theory*, 65(8):4771–4784, August 2019.
- [ANTSV02] Andris Ambainis, Ashwin Nayak, Amnon Ta-Shma, and Umesh Vazirani. Dense quantum coding and quantum finite automata. *Journal of the ACM*, 49(4):496–511, 2002.
- [Bar89] David A. Mix Barrington. Bounded-width polynomial-size branching programs recognize exactly those languages in NC^1 . *Journal of Computer and System Sciences*, 38(1):150–164, 1989.
- [BBC⁺95] Adriano Barenco, Charles H. Bennett, Richard Cleve, David P. DiVincenzo, Norman Margolus, Peter Shor, Tycho Sleator, John A. Smolin, and Harald Weinfurter. Elementary gates for quantum computation. *Physical Review A*, 52(5):3457–3467, 1995. arXiv:quant-ph/9503016.
- [BBC⁺19] Sergey Bravyi, Dan Browne, Padraic Calpin, Earl Campbell, David Gosset, and Mark Howard. Simulation of quantum circuits by low-rank stabilizer decompositions. *Quantum*, 3:181, 2019.
- [BCHK20] Michael Beverland, Earl Campbell, Mark Howard, and Vadym Kliuchnikov. Lower bounds on the non-Clifford resources for quantum computations. *Quantum Science and Technology*, 5(3):035009, May 2020.
- [BCK⁺14] Harry Buhrman, Richard Cleve, Michal Koucký, Bruno Loff, and Florian Speelman. Computing with a full memory: catalytic space. In *Proceedings of the 46th Annual ACM Symposium on Theory of Computing (STOC)*, pages 857–866, 2014.
- [BdW01] Harry Buhrman and Ronald de Wolf. Communication complexity lower bounds by polynomials. *Proceedings of the 16th IEEE Conference on Computational Complexity (CCC)*, pages 120–130, 2001.
- [BFM⁺25] Harry Buhrman, Marten Folkertsma, Ian Mertz, Florian Speelman, Sergii Strelchuk, Sathyawageeswar Subramanian, and Quinten Tupker. Quantum catalytic space, 2025. TQC 2025.
- [BOC92] Michael Ben-Or and Richard Cleve. Computing algebraic formulas using a constant number of registers. *SIAM Journal on Computing*, 21(1):54–58, 1992.
- [BP99] Alex Brodsky and Nicholas Pippenger. Characterizations of 1-way quantum finite automata. Technical Report TR-99-03, University of British Columbia, 1999. arXiv:quant-ph/9903014.
- [BPP00] Joan Boyar, René Peralta, and Denis Pochuev. On the multiplicative complexity of Boolean functions over the basis $(\wedge, \oplus, 1)$. *Theoretical Computer Science*, 235(1):43–57, 2000.
- [BPRS90] László Babai, Pavel Pudlák, Vojtech Rödl, and Endre Szemerédi. Lower bounds to the complexity of symmetric Boolean functions. *Theoretical Computer Science*, 74(3):313–323, 1990.
- [BST90] David A. Mix Barrington, Howard Straubing, and Denis Thérien. Non-uniform automata over groups. *Information and Computation*, 89(2):109–132, 1990.
- [BYM22] Sergey Bravyi, Theodore J. Yoder, and Dmitri Maslov. Efficient ancilla-free reversible and quantum circuits for the Hidden Weighted Bit function. *IEEE Transactions on Computers*, 71(5):1170–1180, 2022. arXiv:2007.05469.
- [DBCM25] Suman Dutta, Anupam Basu Bhaumik, Anupam Chattopadhyay, and Subhamoy Maitra. Optimal T depth quantum circuits for implementing arbitrary Boolean functions. arXiv:2506.01542, 2025.

- [DN06] Christopher M. Dawson and Michael A. Nielsen. The Solovay–Kitaev algorithm. *Quantum Information and Computation*, 6(1):81–95, 2006. arXiv:quant-ph/0505030.
- [Dra00] Thomas G. Draper. Addition on a quantum computer, 2000. arXiv:quant-ph/0008033.
- [FG18] Austin G. Fowler and Craig Gidney. Low overhead quantum computation using lattice surgery. 2018. arXiv:1808.06709.
- [FHW26] Wang Fang, Chris Heunen, and Qisheng Wang. Unitary Synthesis with Near-Optimal T-Count for Near-Clifford Unitaries. arXiv:2607.12907, 2026.
- [Gid17] Craig Gidney. Factoring with $n + 2$ clean qubits and $n - 1$ dirty qubits. arXiv:1706.07884, 2017.
- [Gid18] Craig Gidney. Halving the cost of quantum addition. *Quantum*, 2:74, 2018. arXiv:1709.06648.
- [GIKL23] Sabee Grewal, Vishnu Iyer, William Kretschmer, and Daniel Liang. Low-Stabilizer-Complexity Quantum States Are Not Pseudorandom. In *14th Innovations in Theoretical Computer Science Conference (ITCS 2023)*, 2023.
- [GKW24] David Gosset, Robin Kothari, and Kewen Wu. Quantum state preparation with optimal T-count. arXiv:2411.04790, 2024.
- [GKZ25] David Gosset, Robin Kothari, and Chenyi Zhang. Multi-qubit Toffoli with exponentially fewer T gates. arXiv:2510.07223, 2025.
- [GMPY25] Uma Girish, Alex May, Natalie Parham, and Henry Yuen. Magic and communication complexity. arXiv:2510.07246, 2025.
- [GNW21] David Gross, Sepehr Nezami, and Michael Walter. Schur–Weyl duality for the Clifford group with applications: property testing, a robust Hudson theorem, and de Finetti representations. *Communications in Mathematical Physics*, 385:1325–1393, 2021.
- [GS08] Ben Green and Tom Sanders. A quantitative version of the idempotent theorem in harmonic analysis. *Annals of Mathematics*, 168(3):1025–1054, 2008. arXiv:math/0611286.
- [GS13] Brett Giles and Peter Selinger. Exact synthesis of multiqubit Clifford+T circuits. *Physical Review A*, 87(3):032332, 2013. arXiv:1212.0506.
- [HC17] Mark Howard and Earl Campbell. Application of a Resource Theory for Magic States to Fault-Tolerant Quantum Computing. *Physical Review Letters*, 118(9):090501, 2017. arXiv:1609.07488.
- [HRS16] Thomas Häner, Martin Roetteler, and Krysta M. Svore. Factoring using $2n + 2$ qubits with Toffoli based modular multiplication, 2016. *Quantum Information and Computation* 17(7–8), 673–684 (2017).
- [Jon13] Cody Jones. Low-overhead constructions for the fault-tolerant Toffoli gate. *Physical Review A*, 87(2):022328, 2013. arXiv:1212.5069.
- [JW21] Jiaqing Jiang and Xin Wang. Lower bound for the T count via unitary stabilizer nullity. arXiv:2103.09999, 2021.
- [KASD26] Abhoy Kole, Majd Assaad, Till Schnitka, and Rolf Drechsler. Adaptive Clifford+ t decomposition of large Toffoli gates with one clean ancilla. arXiv:2605.18169, 2026.

- [KG24] Tanuj Khattar and Craig Gidney. Rise of conditionally clean ancillae for efficient quantum circuit constructions, 2024. *Quantum* 9, 1752 (2025).
- [Kre95] Ilan Kremer. Quantum communication. Master’s thesis, The Hebrew University of Jerusalem, 1995.
- [KSD26] Abhoy Kole, Till Schnittka, and Rolf Drechsler. Measurement-driven adaptive low-overhead implementation of multi-controlled Toffoli gates. *arXiv:2605.18159*, 2026.
- [KSS25] Niels Kornerup, Jonathan Sadun, and David Soloveichik. Tight Bounds on the Spooky Pebble Game: Recycling Qubits with Measurements. *Quantum*, 9:1636, 2025. *arXiv:2110.08973*.
- [Lit19] Daniel Litinski. A Game of Surface Codes: Large-Scale Quantum Computing with Lattice Surgery. *Quantum*, 3:128, 2019. *arXiv:1808.02892*.
- [LKS24] Guang Hao Low, Vadym Kliuchnikov, and Luke Schaeffer. Trading T gates for dirty qubits in state preparation and unitary synthesis. *Quantum*, 8:1375, 2024. *arXiv:1812.00954*.
- [LL26] Jingquan Luo and Lvzhou Li. Sparse quantum state preparation with sublinear T-count. *arXiv:2608.00414*, 2026.
- [Mas16] Dmitri Maslov. Advantages of using relative-phase Toffoli gates with an application to multiple control Toffoli optimization. *Physical Review A*, 93(2):022311, 2016.
- [Moo99] Cristopher Moore. Quantum Circuits: Fanout, Parity, and Counting. *arXiv:quant-ph/9903046*, 1999.
- [MS92] Roland Mirwald and Claus-Peter Schnorr. The multiplicative complexity of quadratic Boolean forms. *Theoretical Computer Science*, 102(2):307–328, 1992.
- [MSC⁺19] Giulia Meuli, Mathias Soeken, Earl Campbell, Martin Roetteler, and Giovanni De Micheli. The role of multiplicative complexity in compiling low T-count oracle circuits. In *2019 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, pages 1–8, 2019. *arXiv:1908.01609*.
- [Nay99] Ashwin Nayak. Optimal lower bounds for quantum automata and random access codes. In *40th Annual Symposium on Foundations of Computer Science (FOCS 1999)*, pages 369–376, 1999. *arXiv:quant-ph/9904093*.
- [New91] Ilan Newman. Private vs. common random bits in communication complexity. *Information Processing Letters*, 39(2):67–71, 1991.
- [Nis93] Noam Nisan. The communication complexity of threshold gates. In *Combinatorics, Paul Erdős is Eighty*, volume 1, pages 301–315. János Bolyai Mathematical Society, 1993.
- [NZ26] Junhong Nie and Wei Zi. Nearly optimal quantum circuits for Boolean oracles. *arXiv:2607.28402*, 2026.
- [Par25] Natalie Parham. Quantum circuit lower bounds in the magic hierarchy. *arXiv:2504.19966*, 2025.
- [PPZ92] Michael S. Paterson, Nicholas Pippenger, and Uri Zwick. Optimal carry save networks. In *Boolean Function Complexity*, volume 169 of *London Mathematical Society Lecture Note Series*, pages 174–201. Cambridge University Press, 1992.
- [San19] Tom Sanders. Boolean functions with small spectral norm, revisited. *Mathematical Proceedings of the Cambridge Philosophical Society*, 167(2):335–344, 2019. *arXiv:1804.04050*.

- [San20] Tom Sanders. Bounds in Cohen’s idempotent theorem. *Journal of Fourier Analysis and Applications*, 26(2):25, 2020. arXiv:1610.07092.
- [Sav97] John E. Savage. *Models of Computation: Exploring the Power of Computing*. Addison-Wesley Longman Publishing Co., Inc., USA, 1st edition, 1997.
- [Sel13] Peter Selinger. Quantum circuits of T-depth one. *Physical Review A*, 87(4):042302, 2013. arXiv:1210.0974.
- [SS05] Martin Sauerhoff and Detlef Sieling. Quantum branching programs and space-bounded nonuniform quantum complexity. *Theoretical Computer Science*, 334(1–3):177–225, 2005. arXiv:quant-ph/0403164.
- [TT16] Yasuhiro Takahashi and Seichiro Tani. Power of uninitialized qubits in shallow quantum circuits, 2016. *Theoretical Computer Science* 851, 129–153 (2021).
- [VW04] Farrokh Vatan and Colin Williams. Optimal quantum circuits for general two-qubit gates. *Physical Review A*, 69(3):032315, 2004. arXiv:quant-ph/0308006.
- [WWS19] Xin Wang, Mark M. Wilde, and Yuan Su. Quantifying the magic of quantum channels. *New Journal of Physics*, 21(10):103002, 2019.
- [YA26] Soichiro Yamazaki and Seiseki Akibue. Multi-qubit controlled gate with optimal T -count, 2026. arXiv:2603.14202.
- [ZNS25] Wei Zi, Junhong Nie, and Xiaoming Sun. Shallow Quantum Circuit Implementation of Symmetric Functions with Limited Ancillary Qubits. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 44:3060–3072, 2025. arXiv:2404.06052.
- [ZYNZ26] Wei Zi, Pei Yuan, Junhong Nie, and Shengyu Zhang. Efficient depth–ancilla tradeoffs for Hamming weight computation and symmetric Boolean functions, 2026. arXiv:2608.04627.